2026 Summer Undergraduate Research Fellowship Interim Report II

Task-Driven Neural Geometries for Speech Production

Yourui Shao

Mentors:
Dr. Alessandro Marin Vargas, Neural Prosthetics Translational Lab, Stanford University
Prof. Frank Willett, Neural Prosthetics Translational Lab, Stanford University
Associate Mentor:
Prof. Yisong Yue, Computing and Mathematical Sciences

Background

In our initial hypotheses, we theorized that rotational dynamics—where sequence items move through a queue defined by subspaces previously observed in working memory [1, 2]—emerge in speech representations as a consequence of restrictions on memory during the delay period—that, by applying noise, reducing the hidden state size, or adding low rank adaptors to compress the hidden state between time steps, a perfect rotating queue would emerge; and that the absence of such elements would induce a fixed queue geometry.

What we seem to observe both in our task-driven models and in neural data, however, seems to be more complex (Figs. 3,1). Given an understanding of the output-null and output-potent subspaces, it is now obvious that a perfectly fixed or competitive queue is impossible. However, we observe that the distinction between a fully fixed or competitive queue and fully rotating queue is not binary. Instead, during output, some models do not present phonemes which mechanically rotate into slots occupied by its predecessor in the previous time step perfectly over all time steps. Rather, in those models, rotation is concentrated in the representations of phonemes closest to output, while more distal phonemes lie fixed in slow-moving or constant slots.

We additionally observe some instances of queues that are fully rotational, in which translation into the slot occupied by the previous phoneme dominates over identity for all phonemes. Curiously, the degree to which such rotation occurs seems at first glance independent of singular isolated factors—delay period length, hidden state dimensionality (true or low-rank), weight decay, and sequence length. Thus, we seek instead to understand *what circumstances affect the extent of rotation during output, and why perfect rotation doesn't always occur.*

We determined that the queue behavior during output is partially dependent on delay period hidden state geometry—that training a new model initialized with a hidden state sourced from the delay period of a source model loosely preserves the extent of rotation. As such, we decompose our question into two parts:

1. How do task conditions and training hyperparameters systematically affect the emergent hidden state geometry prior to output?
2. How do hidden state geometries prior to output differ in models that full rotate or partially rotate?

Ultimately, we find that no single factor universally induces full or partial rotations. However, we do find a significant correlation between rotation depth and how "rotation ready" the hidden state is prior to output, measured by how well a single, universal linear transformation can map each phoneme position's subspace into that of the preceding one. We do isolate factors which in-combination and in limited settings induce greater rotation depth. However, we argue that the difficulty of doing so and limitations in applicability suggest that perfect rotational queues may be the exception, limited to short sequences and recall horizons. Conversely, real speech sequence recall and production may be dictated by more independently regulated output trajectories.

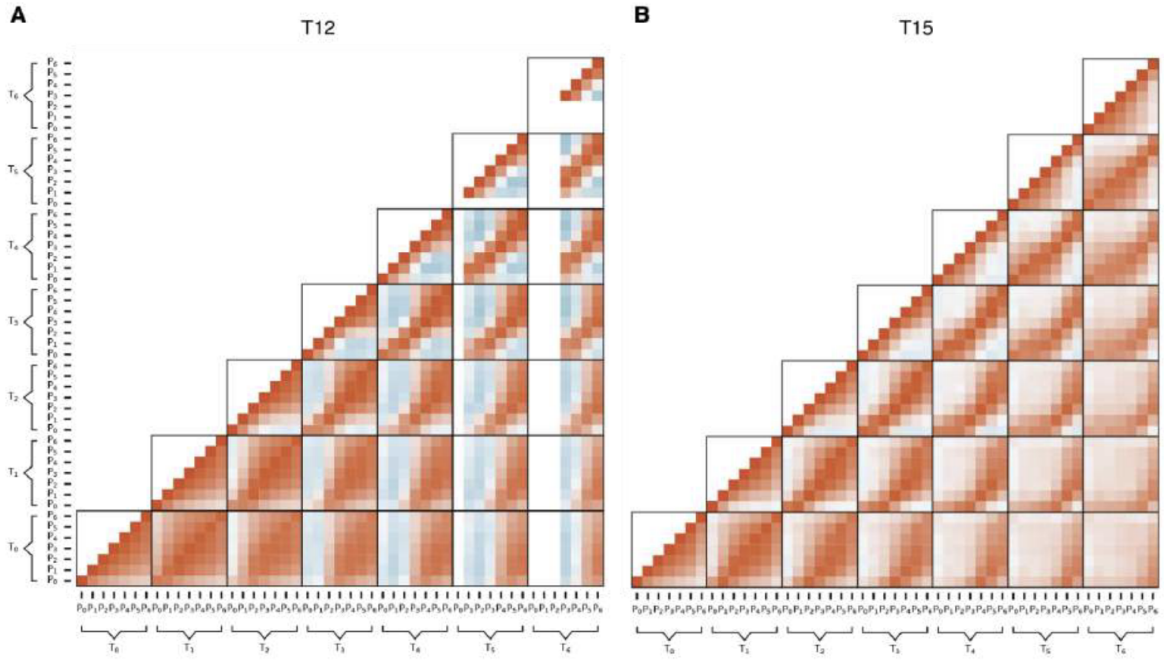


Figure 1: Similarity heatmap for phoneme item representations from fitting SLIM weights on T12 and T15 neural data, sourced from previous work in the group. The persistent diagonal for later positions indicate phoneme positions whose representations are highly similar to that of their own at other time steps, indicating slow or fixed slots for those later positions.

Task Setting

We train Gated Recurrent Units (GRUs), a category of recurrent neural networks (RNNs) on a serial phoneme sequence recall task. Given a phoneme sequence of length l of phonemes $\phi_0, \ldots, \phi_{l-1} \in P \cup \{[\text{sil}]\}$ (where P is the set of 39 English phonemes used), a delay period of length m, and null, go, silent, and stop tokens [null], [go], [sil], [stop] we task the GRU to output:

t	0	…	$l-1$	l	…	$l+m-1$	$l+m$	…	$2l+m-1$	$2l+m$
input	ϕ_0	…	ϕ_{L-1}	[null]	[null]	[null]	[go]	[go]	[go]	[go]
output	[sil]	[sil]	[sil]	[sil]	[sil]	[sil]	ϕ_0	…	ϕ_{l-1}	[stop]

We use one-hot encodings to express phoneme identity during input for each of the 38 phonemes P and one [sil]. We include one additional input channel that is set to 1 during the go period to express [go] and 0 otherwise. [null] is expressed as a 0-vector input. During output, each of the 38 phonemes in P and [sil], [stop] are allotted an output logit.

We apply softmax on the output logits to determine intended output identity during inference. During training, we use a cross-entropy loss objective, but supply an additional mean-squared distance coefficient to minimize and prevent softmax from masking large variations in the raw output logits.

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \text{mean}_{b,t,c} z_{btc}^2$$

where b, t, and c are the batch items, time steps, and logits respectively, and $\lambda = 1.0$ by default.

We obtain our phoneme sequences from the first 10,000 entries the FineWeb dataset [3] using CMUdict [4] and g2p-en [5]. To construct the training set, we filter by a maximum or exact sequence length, truncating at word boundaries to

create sufficient samples for shorter lengths. Unless specified otherwise, during training, we employ a dropout of 0.2, batch size 64, (train, val, test) split of $(0.8, 0.1, 0.1)$, learning rate of 0.001 with the AdamW optimizer and weight decay 10^{-4}, early stopping with patience 20, and an unrestricted number of epochs.

The Geometry of Output-Potent Subspaces Require Rotation

In both neural activity [6], there exists a conceptual readout layer $\mathbf{W}$ applied onto the neural activity to compute the downstream effects (e.g., speech or motion). The analogous in our task-driven RNNs is the final readout layer applied on the hidden state

$$\mathbf{z}_t = \mathbf{W}\mathbf{h}_t + \mathbf{b}$$

when we isolate for phoneme logits (not [sil] or [stop]):

$$\mathbf{z}'_t = \mathbf{W}'\mathbf{h}_t + \mathbf{b}$$

The component of $\mathbf{h}_t$ that lies in the null space of $\mathbf{W}'$, Ker $\mathbf{W}'$, has no effect on the value of $\mathbf{z}'_t$. As $\mathbf{z}'_t \in \mathbb{R}^{39}$ and $\mathbf{h}_t$ is generally of a higher dimension, we find that orthogonal complement, the output-potent row space of $\mathbf{W}'$, Ran $\mathbf{W}'^{\mathsf{T}}$, has maximal dimension rank $\mathbf{W}'$, which is maximally 41. Thus, only a small subspace of the hidden state space has any effect on output. Namely, for a hidden state $\mathbf{h}_t$, only the output-potent component $\text{proj}_{\text{Ran } \mathbf{W}'^{\mathsf{T}}} \mathbf{h}_t$ affects the value of $\mathbf{z}'_t$.

We can imagine that $\text{proj}_{\text{Ran } \mathbf{W}'^{\mathsf{T}}} \mathbf{h}_t$ must encode at minimum information about the output phoneme ϕ_t for output time step t. Specifically, $\max_c \mathbf{z}'_t$ encodes ϕ_t. Note that $\mathbf{W}'$ is fixed for all time steps. For a fixed queue conception, as the representation of individual phonemes in $\mathbf{h}_t$ lie in static subspaces ("slots"), we imagine that the output-potent space $\text{proj}_{\text{Ran } \mathbf{W}'^{\mathsf{T}}} \mathbf{h}_t$ must then encode all phonemes for the given sample simultaneously. This behavior, introducing insurmountable collisions in representations, is evidently disadvantageous compared to utilizing at least some portion of Ker $\mathbf{W}'$ to store phonemes which are not currently expressed. Then, the representations for those phonemes can "rotate" into the output-potent space Ran $\mathbf{W}'^{\mathsf{T}}$ on their turn for expression (Fig. 2).

Note that this formulation is independent of the delay period length, input format, and (largely) the size of $\mathbf{h}$ chosen. Thus, we conclude that the emergence of a fixed queue during output is theoretically impossible, though it could and often does govern input and delay period dynamics. We also observe this empirically. We describe our method for decomposing $\mathbf{h}$ into constituent phoneme representations and identifying the output-null and output-potent components in Interim Report I. In all instances, we observe that phoneme at their time of output all occupy a highly similar subspace, with generally similar preceding subspace trajectories taken to reach that subspace.

In addition to a shared "output slot" which all phonemes must ultimately rotate into, we also observe the presence of a shared "input slot" (Fig 4). All phonemes, when they are introduced during serial input, are inserted into the hidden state in the same position and must later be rotated out and processed according to the task. This is consistent with the identification of a temporary space sequence items occupy prior to reorganization in macaque working memory studies [7].

Quantifying Rotation

We have observed that a slow or fixed queue dominates for phonemes at positions further from expression during output time steps, with phoneme representations more similar to that of their own position at the previous time step than that of the preceding phoneme at the previous time step (Figs. 3,**??**). We refer to the output-nearing rotation queue with fixed or slow representations otherwise as *partially rotating*. For these partial rotations, how far back in the queue a phoneme begins to rotate and how distinct the subspaces of adjacent phonemes are are also seem to vary between model variants.

We also developed an alternative phoneme recall formulation, in which the delay period is defined not relative to the end, but to the start of the phoneme input period. For an "on-input" delay period length of m, the timeline is as follows:

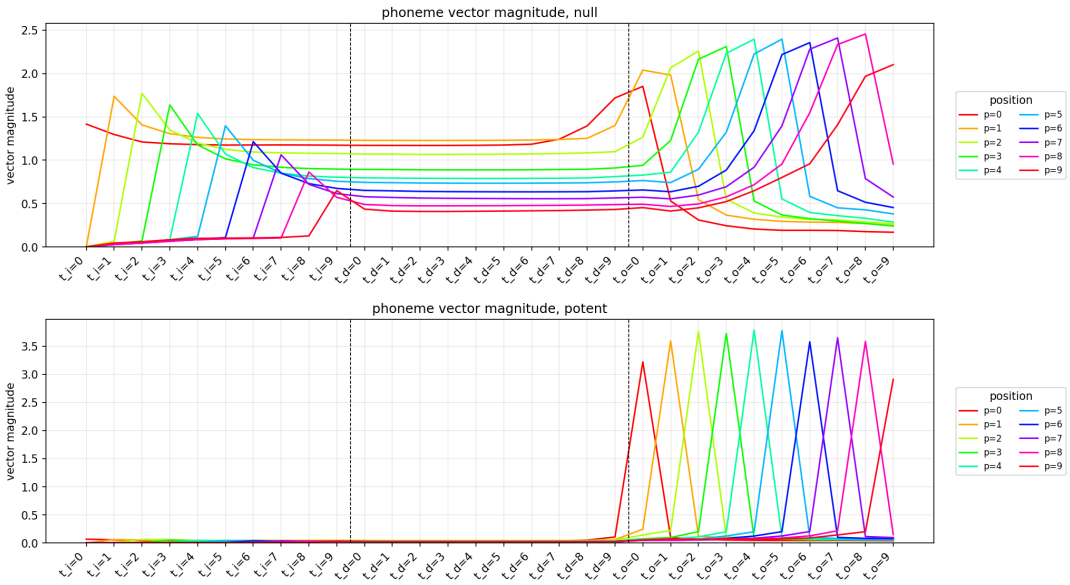


Figure 2: We separately fit a new set of SLIM weights for the null and potent spaces at each time step, and report the magnitude of each phoneme's representation in that hidden state. In the null space, each phoneme is encoded between its input and output. In the potent space, phonemes are encoded only at the time step at which they are expressed. Run performed with $l \leq 10$, $m = 10$, hidden state size $d = 256$.

t	0	…	m	…	$m+l-1$	$m+l$
input	ϕ_0	…	ϕ_m + [go]	…	[null]	[null]
output	[sil]	[sil]	ϕ_0	…	ϕ_{l-1}	[stop]

We observe that this task consistently induces *fully rotating* geometry—a subspace is assigned for each position relative to expression and phoneme identities rotate through those subspaces (Fig. 6) This is comparable to the continuous streaming task of previous working memory sequence representation studies, in which N-back recall tasks are theorized to induce a fully rotating geometry in working memory [1].

We thus seek to quantify the extent of rotation and subspace overlap with a few parameters. Let Δ, for a given position relative to output p_{rel}, denote the difference between the extent that a phoneme representation is rotating versus fixed. When $\Delta > 0$, the item at the given position is rotating, and when $\Delta < 0$, the item is fixed. For SLIM projections $\mathbf{B}_{t,p} \in \mathbb{R}^{d\times 40}$, assigning indices $i = 0, 1, ..., 39$ where $|P \cup \{[\text{sil}]| = 40$, and cosine similarity function sim,

$$\Delta(p_{\text{rel}}) = \underset{t:\, p-t=p_{\text{rel}}}{\text{mean}} \; \underset{i}{\text{mean}} \left(\text{sim}\left(\mathbf{B}_{t,p}\mathbf{e}_i, \mathbf{B}_{t+1,p+1}\mathbf{e}_i\right) - \text{sim}\left(\mathbf{B}_{t,p}\mathbf{e}_i, \mathbf{B}_{t+1,p}\mathbf{e}_i\right)\right)$$

A p_{rel} captures all phoneme representations at position $p = t + p_{\text{rel}}$ for output time step t (e.g., $p_{\text{rel}} = 1$ includes (t, p) pairs $(0,1), (1,2), (2,3), \ldots$). Rotation depth $\pi_{\text{rot}} \leq l - 1$ is then the maximum p_{rel} such that $\Delta < 0$ for all $p_{\text{rel}} \leq \pi_{\text{rot}}$ (Fig. 12). When $\pi_{\text{rot}} = l - 1$, we consider the model to be "fully" or "perfectly" rotating. We also measure the hidden state subspace overlap of adjacent phonemes for the rotating and fixed portions:

$$s_{\text{rot}} = \underset{\substack{t,p:\, p-t<\pi_{\text{rot}} \\ p+1-t<\pi_{\text{rot}}}}{\text{mean}} \; \underset{i}{\text{mean}} \; \text{sim}\left(\mathbf{B}_{t,p}\mathbf{e}_i,\, \mathbf{B}_{t,p+1}\mathbf{e}_i\right)$$

$$s_{\text{fixed}} = \underset{\substack{t,p:\, p-t\geq\pi_{\text{rot}} \\ p+1-t\geq\pi_{\text{rot}}}}{\text{mean}} \; \underset{i}{\text{mean}} \; \text{sim}\left(\mathbf{B}_{t,p}\mathbf{e}_i,\, \mathbf{B}_{t,p+1}\mathbf{e}_i\right)$$

A higher s indicates greater overlap and less orthogonality between adjacent slot subspaces (Fig. 12).

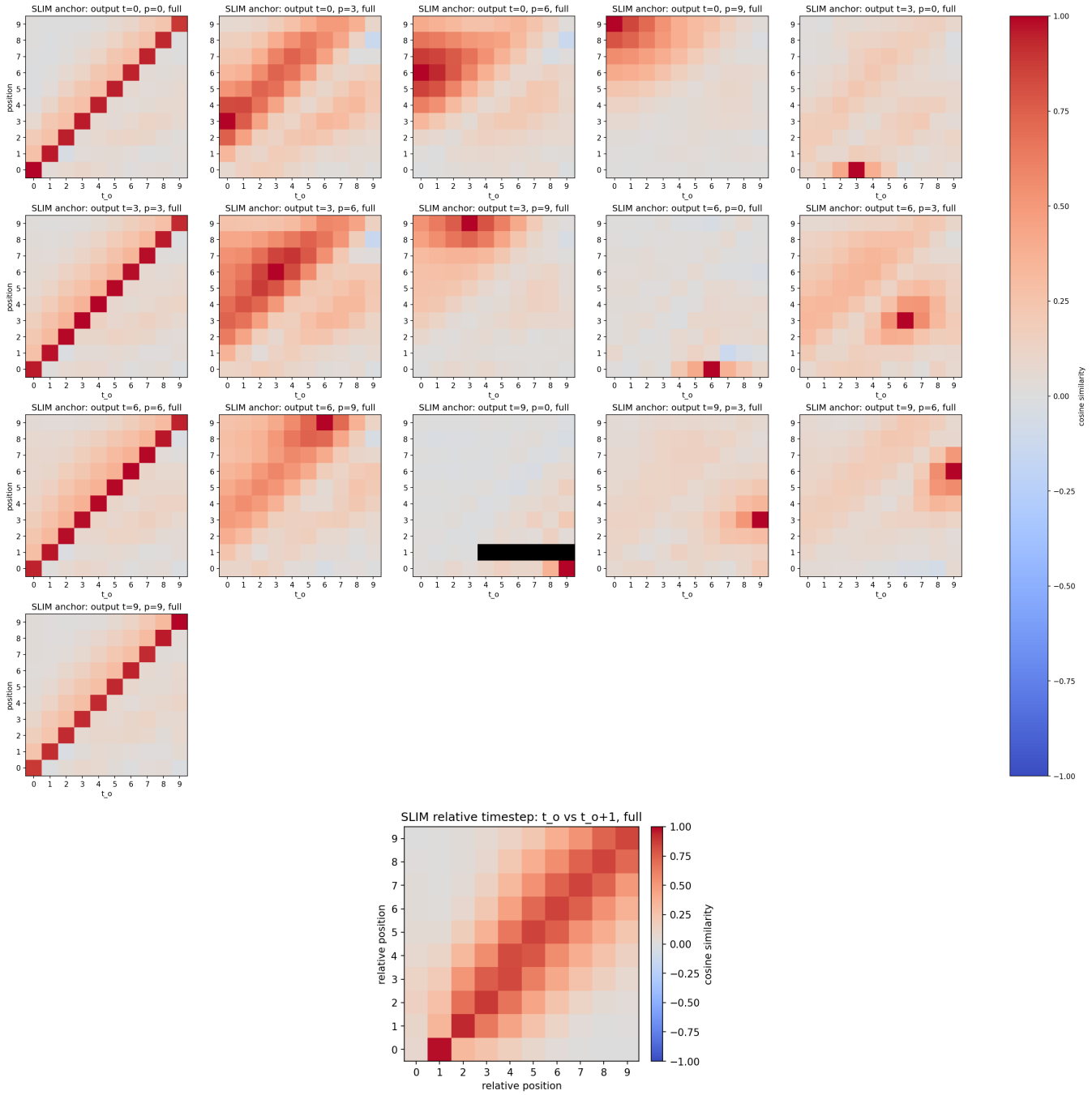


Figure 3: Output period phoneme representations for a $l \leq 10$, $m = 10$, $d = 256$ run. **Above**: Cosine similarities between phoneme representations for various positions and output period time steps and anchors, averaged across phoneme identities. For phonemes that are distal to expression (e.g., $p = 6$ at $t = 0$), their representations tend to be closer to that for the same position at neighboring time steps, than that for the preceding or following positions. This indicates that representations are stored in slow-moving or fixed slots. However, output time step representations show high similarity to each other. **Below**: Cosine similarities between phoneme identities in subspaces for position relative to output between a time t and $t + 1$, output period. A rightward shift of 1 position from the diagonal indicates that representations are advancing—"rotating"—in subspaces, taking on the representation of the preceding phoneme at the previous time step. The unmoved diagonal for higher relative positions indicates the lack of such a shift for phonemes further relatively from output.

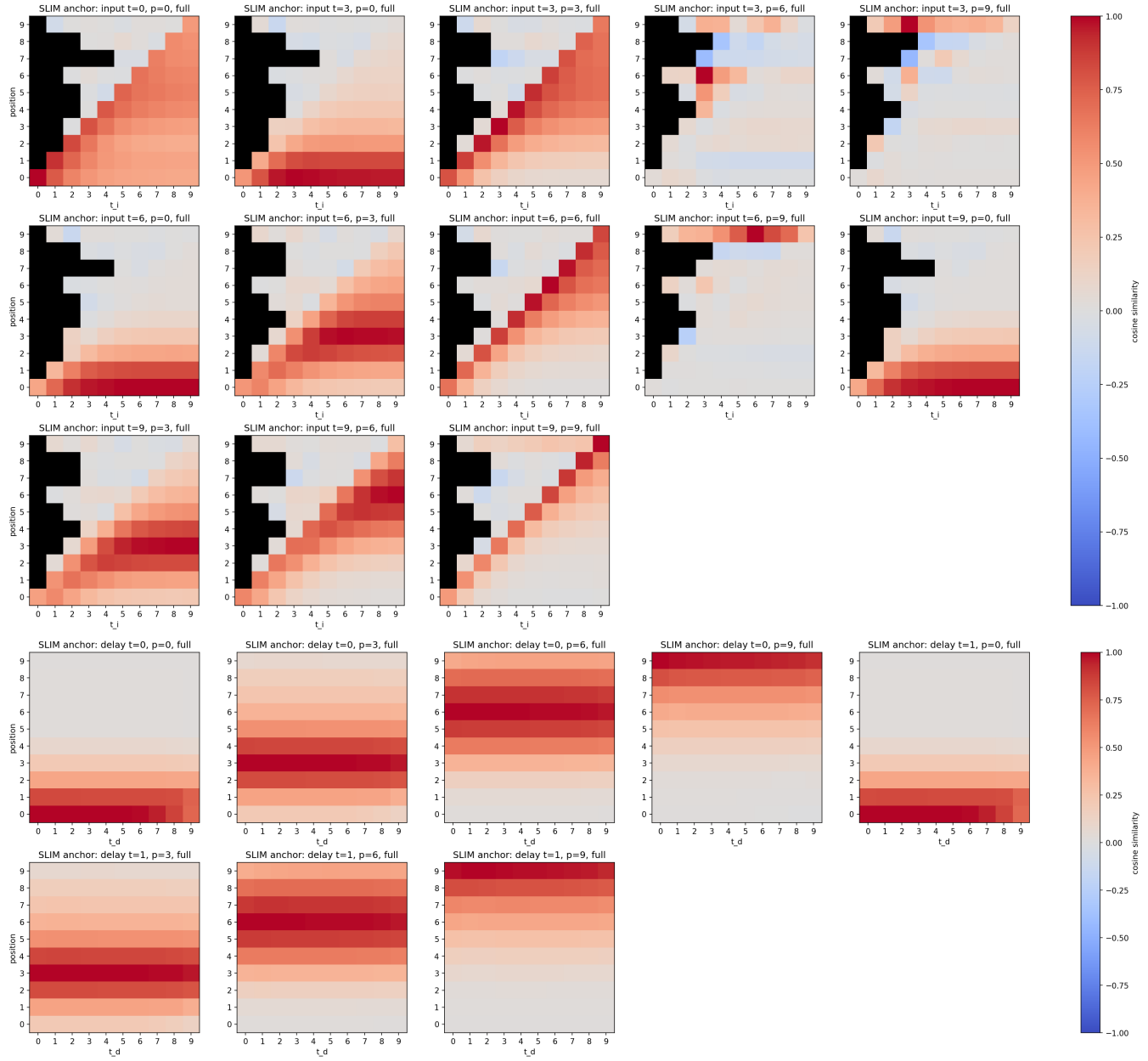


Figure 4: Input (top) and delay (bottom) period phoneme representations for an $l \leq 10$, $m = 10$, $d = 256$ run, displaying cosine similarities between phoneme representations for various positions and output period time steps and anchors, averaged across phoneme identities. During the delay period, representations lie largely in fixed slots with near-constant representations and some subspace overlap with adjacent phonemes.

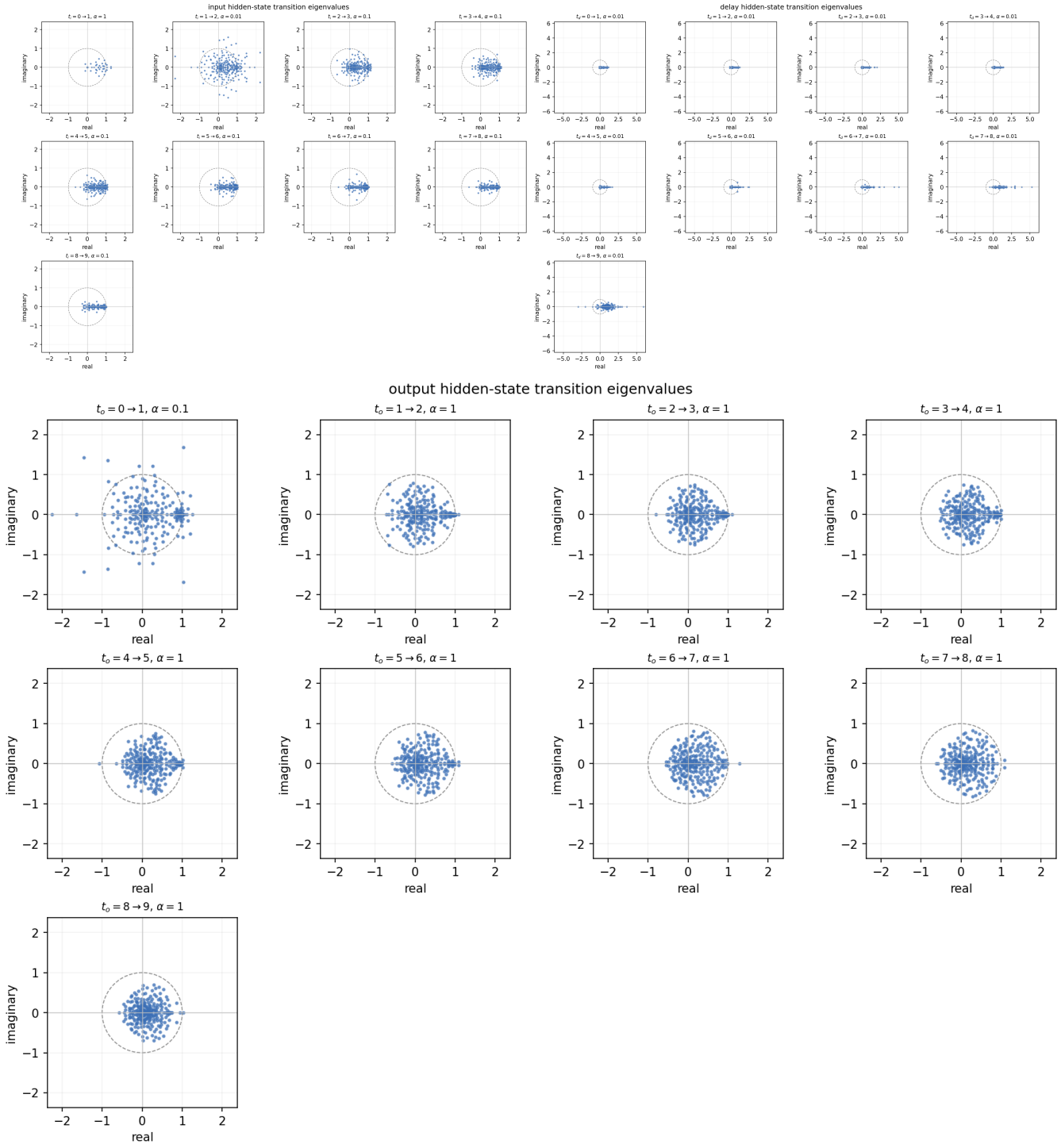


Figure 5: For a $l \leq 10$, $m = 10$, $d = 256$ model, we fitted linear maps to transform the hidden states from one time step to the next for the input (top left), delay (top right), and output (bottom) period. We computed the eigenvalues for the linear maps of each step transition. Eigenvalues near 1 indicate basis vectors that are approximately preserved between time steps, while eigenvalues with high imaginary compnoents indicate basis vectors that are rotated. Delay period eigenvalues are dominated by identity, while input and output time steps are a combination of rotation and identity.

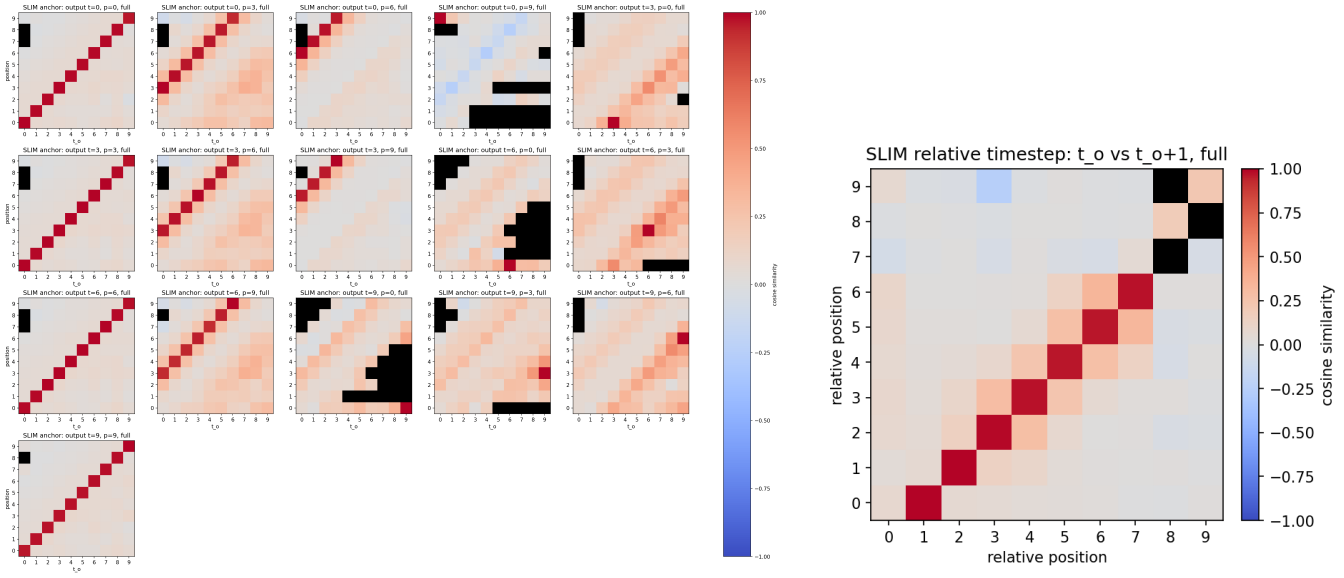


Figure 6: Output period phoneme representations for a $l \leq 10$, on input $m = 6$, $d = 256$ run. See Fig. 3 for comparison. **Left**: Even phonemes that are distal from expression undergo rotation, with representations similar to that of the preceding phoneme in the previous time step and following phoneme in the next time step. **Right**: The rightward one step shift persists for all positions which are decodable given the delay scheme, indicating a full rotation with depth 7.

We can also consider the transition between slow or fixed slots representations to rotating slots near expression to be a measurement of speed—the angular distance between representations between time steps (Fig. 7). Fixed and slow moving representations have low angular speeds, while rotating representations tend to have higher angular speeds.

Output Behavior Is Not Only a Function of Previous Hidden States

We use hypothesize that the rotational geometry during sequence output is strongly dependent on the organization of the hidden state immediately prior, and not strongly affected by the computational solution of the GRU developed to separately handle the input and delay periods. From literature, we recognize that working memory can reorganize itself according to the task without need for developing separate downstream computational strategies. In monkeys, the working memory organized sequence items by *output order* for both forward and reverse recall tasks [7].

To test this hypothesis, we initialize fresh GRUs with the pre-output hidden states of trained RNNs as $\mathbf{h}_0$. Specifically, we used models with sequence length $l = 5, l \leq 5$, delay length $m = 0, 2, 4, 6$, hidden state size $d = 256$, which demonstrate existing variation in rotation depth. For each model, we sampled the hidden state from the last delay time step and used that as $\mathbf{h}_0$ in initializing 3 fresh GRUs with different pairs of different dataset split and training random seeds. We then, beginning on the first time step, require serial output of the original phoneme sequence. We used the same training configurations as the original models (early stopping with patience 10 and min delta 10^{-3}). This thus emulates how the hidden state would be processed if there were no constraints on the model weights from accommodating input and delay computation.

We ask how aligned would the output-time geometry and behavior be with that of the original model. We measure alignment by the conservation of rotation depth; and whether the slot subspaces for phoneme tokens are linearly transformable to that of the original model with higher test-time correlation than to that of other models.

Curiously, we witness a remarkable shift to deeper rotation depth compared to the original models. Of the 6 models with original rotation depth 2, 17 of the 18 $\mathbf{h}_0$ initialized derivative models developed a rotation depth of 3. Similarly, of the 8 original models with rotation depth 3, 21 of 24 derivative models developed a rotation depth of 4. For 10 original models with rotation depth 4, 6 of 30 derivatives regressed to a rotation depth of 3 (Fig. 8). Notably,

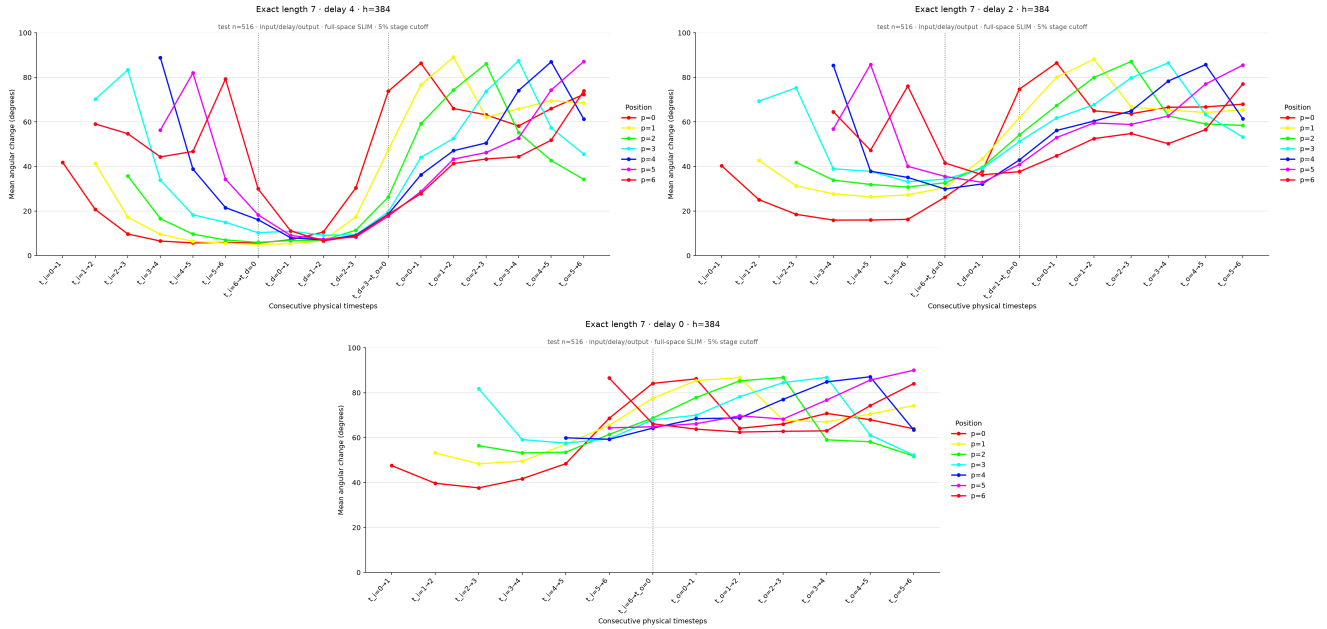


Figure 7: Average speed (change in angle between timesteps) over time for the SLIM representation of a given position for three $l = 7, d = 384$ runs with $m = 0, 2, 4$.

the overall deepening in derivative models does not guarantee ultimately full rotations. We can thus imagine that during real sequence recall, the constraint of having to manage sequences during input, delay, and output affects the mechanism for rotation; the pre-output organization alone doesn't account for all output-stage geometry.

However, we also observe with the $\mathbf{h}_0$ initialized trials some dependency of the output-period geometry on $\mathbf{h}_0$; foremost, no original rotation depth 2 model attains a final rotation depth of 4 in the $\mathbf{h}_0$ initialized dependencies. Additionally, the output-period subspace organizations of dependency models are on average more similar to that of their original models than any other variants. We use a universal linear map to translate all SLIM encoding weights of one model to another, and measure its R^2 performance on held-out time steps position pairs. For output-stage SLIM coefficients of one model $\{\mathbf{B}_{t=0,p=0}, \mathbf{B}_{t=0,p=1}, \ldots, \mathbf{B}_{t=l,p=l}\}$ and that of another $\{\mathbf{B}'_{t=0,p=0}, \mathbf{B}'_{t=0,p=1}, \ldots, \mathbf{B}'_{t=l,p=l}\}$, we arbitrarily sample four in five (t, p) pairs and fit a linear model to map the flattened selection of $\mathbf{B}$ to their equivalent $\mathbf{B}'$ with mean squared error, and evaluate the R^2 of the fit on the remaining flattened weights. On average, the similarity of an $\mathbf{h}_0$ initialized model to its source model attains a mean $R^2 = 0.686$, compared to a mean similarity of $R^2 = 0.540$ to other non-source models ($p = 1.19 \times 10^{-7}$, two-sided). Furthermore, while only 1 of the 24 compared models is the source model to the $\mathbf{h}_0$ initialized derivative, the source model is the most similar under R^2 27.8% of the time. When limiting this comparison to models with the same training configuration as the source model but different dataset split and training random seeds, the distinction is still significant, with mean $R^2 = 0.686$ for source compared to $R^2 = 0.556$ for alternative seeds ($p = 5.95 \times 10^{-7}$, two-sided) and the source models most similar 98.6% of the time. Thus, the structure of the pre-output hidden state still likely impacts the output geometry, and we will qualify this relationship next.

Output Behavior is Still a Function of Previous Hidden States

What in particular about the pre-output hidden state discriminates between different degress of rotation? Perhaps, for full-rotation models, there exists a universal linear map $\mathbf{T}$ can translate the phoneme identity hidden state subspaces for each position and into that for the preceding position. If there exists such a $\mathbf{T}$, then such a transformation can be repeated over output time steps to effectively rotate phonemes towards the front of the queue (Fig. 10). For each SLIM coefficient for the last time step before output t and position p, $\mathbf{B}_{t,p} \in \mathbb{R}^{d \times 41}$, we extract the individual columns for each phoneme identity ϕ and obtain $\mathbf{b}_{t,p,\phi} \in \mathbb{R}^d$ (d is the hidden state size). We thus seek to fit a $\mathbf{T} \in \mathbb{R}^{d \times d}$ such

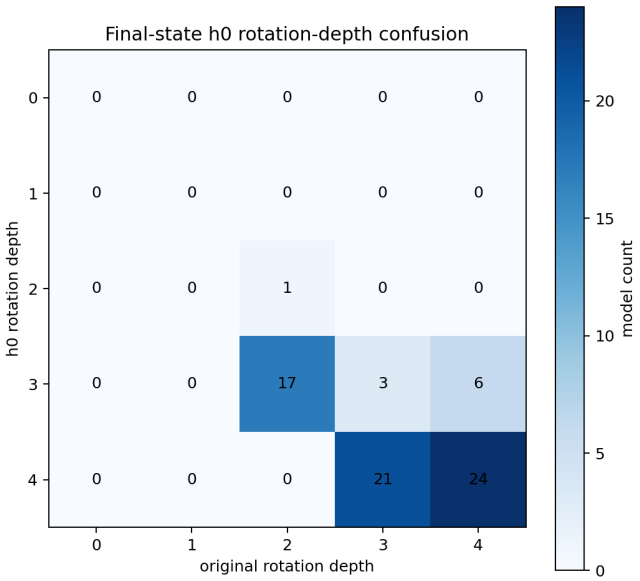


Figure 8: Rotation depths of the original models ($l = 5, l \leq 5$, $m = 0, 2, 4, 6$, $d = 256$) versus the $\mathbf{h}_0$ initialized derivative models.

that

$$\mathbf{T}\mathbf{b}_{t,p,\phi} = \mathbf{b}_{t,p-1,\phi}$$

for all $p > 0$ and ϕ. We find $\mathbf{T}$ using mean-squared error and evaluate $\mathbf{T}$ on a held-out fifth of p, ϕ values, obtaining an R^2 of how well $\mathbf{T}$ generalizes. A high R^2 indicates a good fit. We refer to this property as rotation-readiness and $\mathbf{T}$ as the common map.

Among many sets of runs, the R^2 of $\mathbf{T}$ correlates well with rotation depth. For $l = 7, l \leq 7$ runs (119-211 in the Table of Runs), better $\mathbf{T}$ trends with greater rotation depth (Fig. 9, $p = 0.0010$). Comparing strictly between full and partial runs also yields significant results ($p = 0.0213$, two-sided t-test). However, no significance was found between full and partial runs for the $l = 5, l \leq 5$ runs (rows 83-109, 111, 113, 115 in the Table of Runs, $p = 0.0845$, two-sided t-test), which may be due to the smaller sample sizes. On a more comprehensive set of models with various sequence lengths and delay periods (rows 212-499), we do find that the R^2 of $\mathbf{T}$ differed significantly between full and partial rotations ($p = 1.21 \times 10^{-11}$, two-sided t-test). Ultimately, we conclude that there is a significant relationship between the common-map R^2 score and rotation depth, indicative of the relationship between pre-output hidden state geometry and output-stage behavior.

Rotation Depth Is Not Universally Governed by a Singular Task or Model Condition

Initially, our observations of full rotation depth were restricted to tasks with on-input delays with the alternative timeline formulation (Fig. 6). For such a task, the number of phonemes that must be concurrently represented in the hidden state $L = \min(m + 1, l)$ for a delay period of m, is typically lower than with after-input delay variants, in which all $L = l$ tokens must eventually be represented.

Thus, we initially believed that rotation depth is affected by the hidden state dimensionality—that, if a unique 40 or 41-dimensional subspace ($|P \cup \{[\text{sil}]\}| = 40$, $|P \cup \{[\text{sil}], [\text{stop}]\}| = 41$) was allotted to each ϕ_i and $\mathbf{h} \in \mathbb{R}^{\geq(41\times L)}$, we would be observe fully rotational dynamics. We could also plausibly add a few more subspaces specifically for input or output ($\mathbf{h} \in \mathbb{R}^{\geq(41\times(L+2))}$,). Nevertheless, rotation depth should theoretically scale with hidden state dimension size (d).

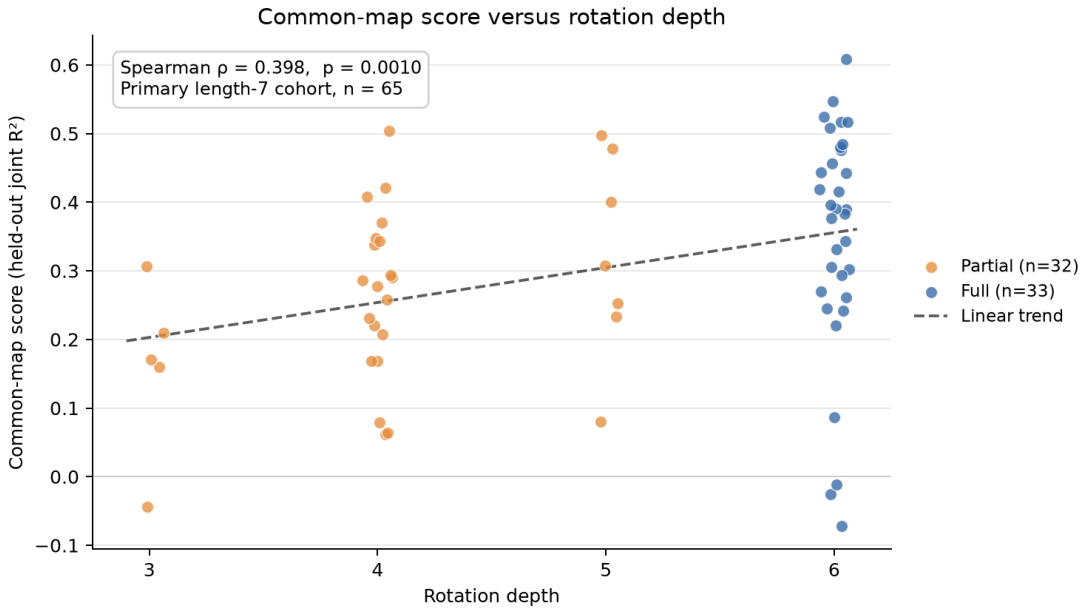


Figure 9: Common-map **T** R^2 score versus rotation depth for $l = 7, l \leq 7$ varieties (runs 119-211). Deeper rotations correlate significantly with better **T** fits.

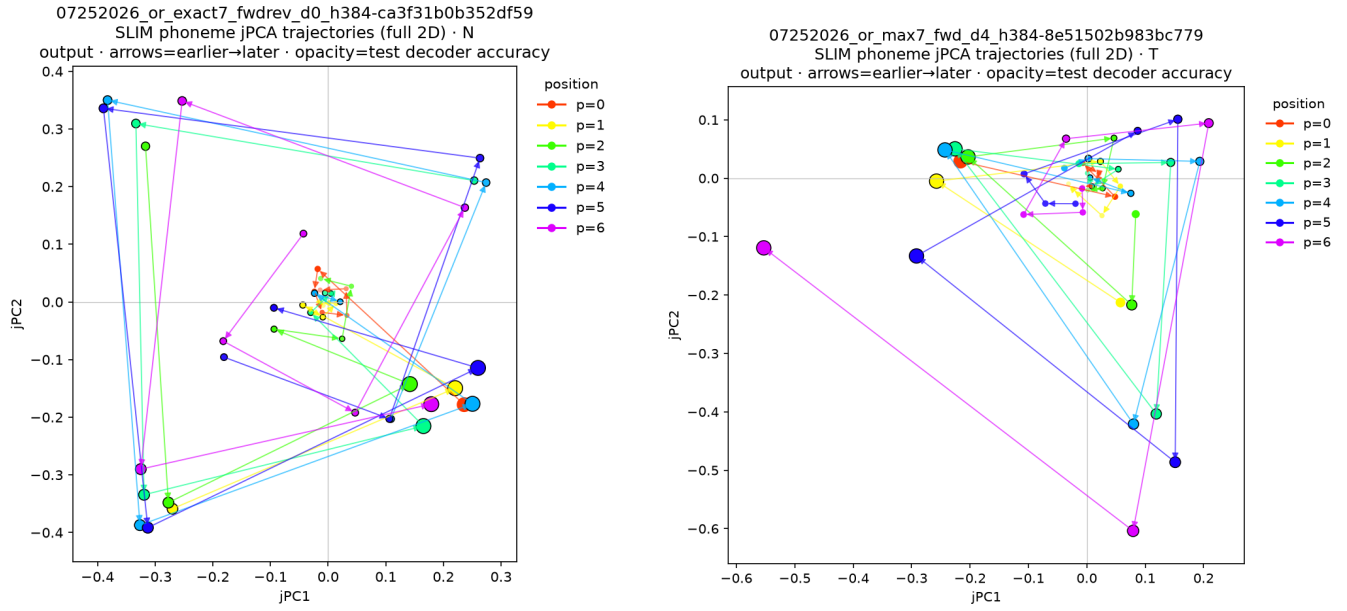


Figure 10: Output-stage SLIM projection representations of individual sequence tokens for phoneme identity N and T, respectively. **Left**: $l = 7, m = 0, d = 384$ trained on both forward and reverse sequence recall tasks, attaining full rotation depth ($\pi_{\text{rot}} = 6$). Tokens generally undertake similar trajectories, with slightly differing magnitudes and small angle deviations. **Right**: $l \leq 7, m = 4, d = 384$ trained on only forward sequence recall, attaining partial rotations depth ($\pi_{\text{rot}} = 4$). Tokens undertake similar but distinct trajectories towards the output subspace (top left corner), and representation magnitudes appear to increase for later phonemes. Partial rotations in general correspond to output-stage trajectories that are less uniform across different tokens.

Additionally, we believed that the *effective* hidden state dimension size—the true number of dimensions, can be affected by other hyperparameters such as weight decay or whether we are representing a true phoneme sequence or a randomly generated one.

Hypothesis 1: Rotation depth is a function of the ratio between effective hidden state dimension, dependent on model hyperparameters, and the number of phoneme units that must be held simultaneously in memory.

To test this hypothesis, we trained a selection of models with hidden state sizes spanning an interval from $\geq (41 \times L)$ to $< (41 \times L)$, with hidden state sizes $512, 448, 384, 320, 256, 128$, and 64 and maximum sequence length 10 samples.

We introduce a variety of ways to quantify the effective dimensionality of the hidden states, d_{eff}. Initially, we used the participation ratio (following Johnston and Fusi [8]) and effective rank. However, we found that as both measures used a threshold to suppress the contribution of smaller activations, which became more infrequent and decreased the effective dimensionality when pooling across a larger sample set and over multiple time steps. Hence, we instead examined PCs obtained from principle component analysis (PCA), asking how many PCs can explain 95% of variability in hidden states over samples. We computed PCs by pooling $\mathbf{h}$ across input, delay, and output time steps, sampling the maximum when stratified by sequence lengths in the test set ($d_{\text{eff}}^{\max}$).

First, we observe that, even when the hidden state is sufficiently large to assign a unique subspace to each sequence item before input and output, where $\mathbf{h} \in \mathbb{R}^{>(41\times L)}$, $d_{\text{eff}}^{\max}$ is much less than $41 \times L$. Indeed, the maximally observed ratio $d_{\text{eff}}^{\max}/L$ is $85:3$. This indicates that heavily shared or overlapping subspaces may be a naturalistic solution rather than an effect of memory constraints.

Furthermore, varying hidden state dimensions had no effect on rotation depth (see Table of Runs). While 512 is greater than 10×41 and also $(10+2) \times 41$ and thus sufficient to accommodate all phoneme items in unique subspaces, there remains significant nonorthogonality in the subspaces of neighboring phonemes ($s_{\text{rot}}, s_{\text{fixed}} >> 0$). When decreasing hidden state sizes to lower values, the rotation depth remained constant at 4, except for hidden state size 64, which reported a rotation depth of 8. However, this was inconclusive as the phoneme error rate (PER) for that run was substantially higher (0.176).

Finally, higher weight decay values, while marginally lowering effective dimensionality, do not appear to have a significant effect on rotation depth (Table 1). We also developed alternative datasets in which we progressively remove the statistical patterns in the real FineWeb-sourced phoneme sequences: scrambling phonemes within word boundaries, scrambling all tokens (phonemes or [sil]) in each sentence, and then randomly selecting the token for each position, with adherence to dataset sentence length distributions. Hypothetically, statistical patterns (e.g., the tendency for consonant and vowel sounds to alternate) could allow for greater representation compression. We observe no drastic effects on rotation depths and did not thoroughly investigate effects on $d_{\text{eff}}^{\max}$ (Fig. 11).

weight decay	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	rotation	s_{rot}	s_{fixed}
0	0.001	145	14.5	4	partial	0.518	0.754
10^{-4}	0.001	145	14.5	4	partial	0.532	0.772
10^{-3}	0.001	138	13.8	4	partial	0.534	0.786
10^{-2}	0	133	13.3	4	partial	0.558	0.779
10^{-1}	0	124	12.4	4	partial	0.537	0.594

Table 1: Rotation metrics for a series of $l = 10, m = 10, d = 512$ runs with varying weight decays.

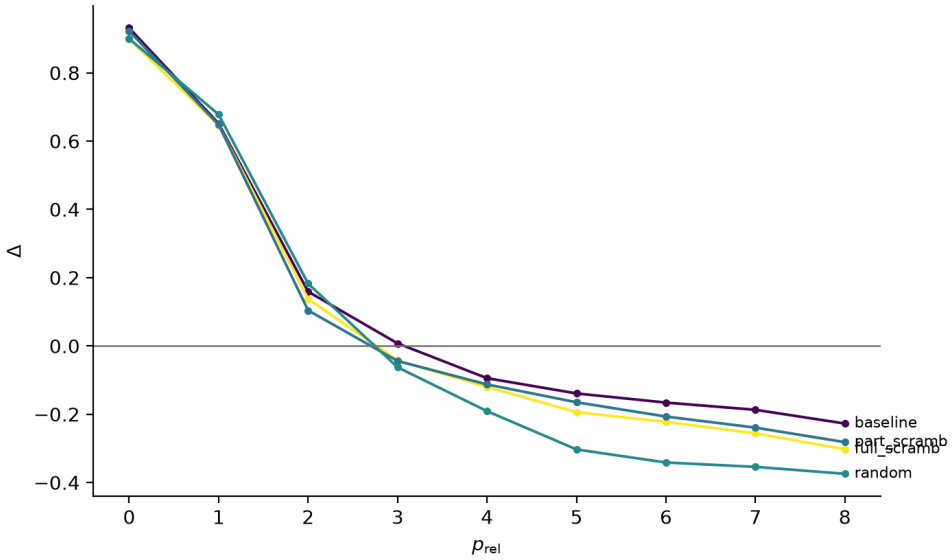


Figure 11: Rotation Δ values over p_{rel} for runs with $l \leq 10, m = 10, d = 512$ with the normal FineWeb dataset, scrambling within words, scrambling across word boundaries, and randomized phonemes.

Hypothesis 2: Perfect rotations occur when the duration a phoneme must be held in memory for is constant across training samples.

While we were unable to identify a significant, universal relationship between effective and allotted hidden state dimensionality with L and the degree of rotation, we did note the distinct behavior of on-input delay formulations consistent with previous working memory literature [1]. For these models, the tokens that must be represented in memory are "streaming." For most time steps, for every token that is output or can be forgotten, a new token must be stored and remembered. Thus, we theorize that knowing in advance how many tokens must be stored and for how long allows the model to organize its hidden state in a manner condusive to full rotations.

In order to test this hypothesis, we construct an another task for which how long a phoneme must be held constantly in memory is certain—training only on fixed length sequences. For sequences of length l and a normal post-input delay period of length m, the number of time steps each phoneme must be retained in the hidden state is $l + m + 1$. When the RNN is trained on sequences of varying lengths, the system on first input does not know l and thus $l + m + 1$ in advance. Thus, we theorize that it may be easier to organize representations if an RNN would be able to anticipate $l + m + 1$. We thus ask: Does training on fixed sequence lengths result in deeper rotations than variable sequence lengths?

Furthermore, we ask if full rotation depth persists in on-input delay, streaming scenarios, if we lengthen m. In exact length sequences, the on-input delay m becomes equivalent to post-input delay $m - l$ when $m \geq l$. As such, we interest ourselves in whether full rotations would continue in such a case.

Finally, we ask that whether effective dimensionality is a function of how long each phoneme must be stored for: If such is known, then if there is a unique slot for each time step that the item is stored, then speculatively the phonemes could advance through all such slots from the input to the output slot. This circumvents the necessity of holding phoneme representations fixed during the delay period—a pattern we commonly observe, which may make less convenient "restarting" rotations during output. Let T denote the said duration, with $T = l + m + 1$ for after-input delays and $T = m + 1$ for on-input delays. Then, under this hypothesis, we will observe full rotations if $\mathbf{h} \in \mathbb{R}^{\geq T \times 41}$ and not necessarily when $\mathbf{h} \in \mathbb{R}^{<T \times 41}$.

We train a family of models on exact and maximum length 5 tasks with after-input delays ranging from 0 to 8. We trained 3 versions of each variant with different dataset splits and training random seeds and averaged the depth across samples. Curiously, we observe that, contrary to our hypothesis, the maximum length trials attained consistently full rotations while exact length trials did not for various delay lengths (Fig. 13). Furthermore, we also

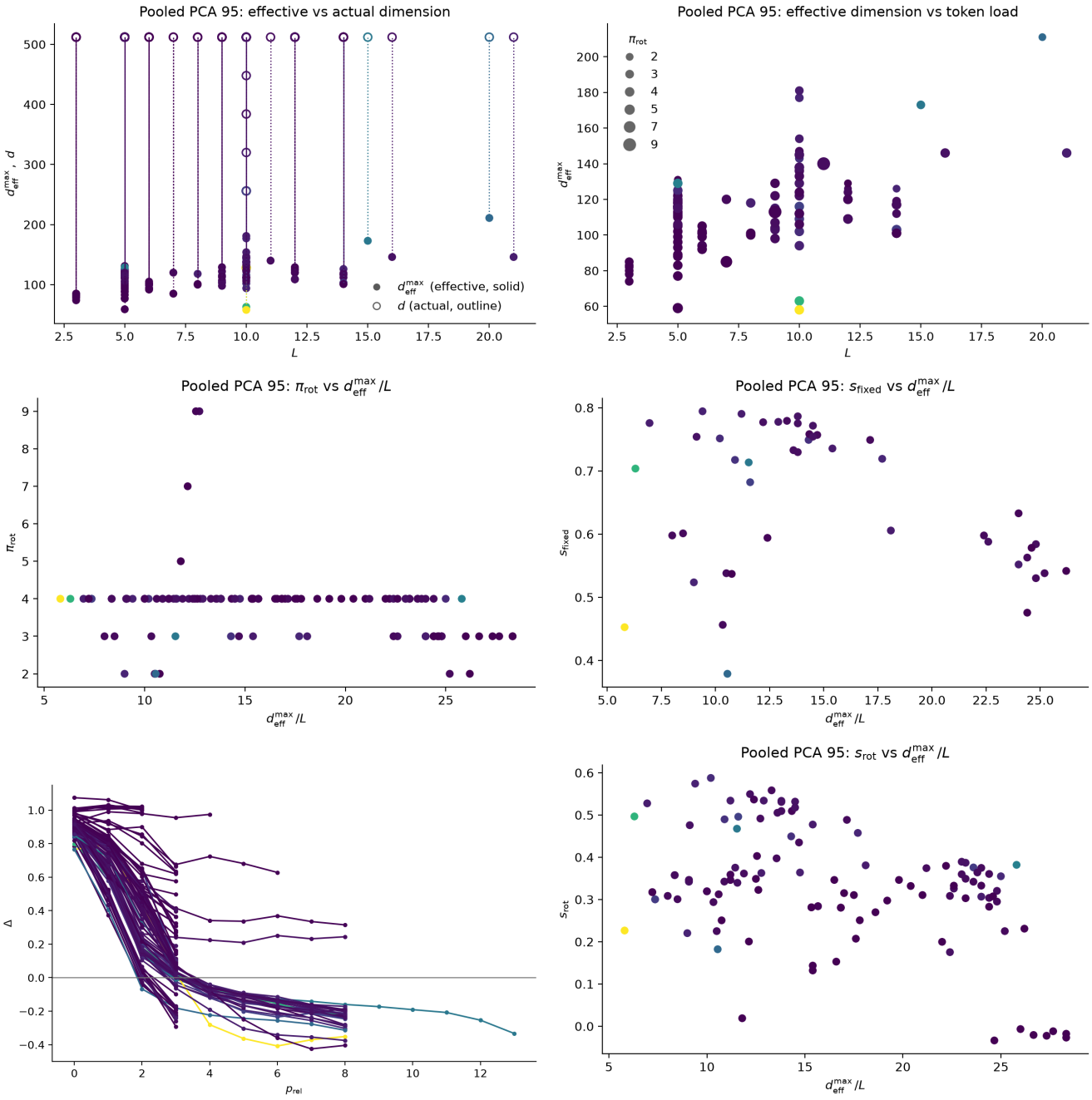


Figure 12: For a collection of runs, d and $d_{\text{eff}}^{\max}$ versus L; π_{rot}, s_{rot}, and s_{fixed} versus $d_{\text{eff}}^{\max}/L$; and Δ over p_{rel}. $d_{\text{eff}}^{\max}$ falls dramatically below d for all runs. A slight positive trend can be observed between L and $d_{\text{eff}}^{\max}$, but all values for $d_{\text{eff}}^{\max}$ lie significantly below $41 \times L$. A weak negative trend can be observed between s_{fixed} and s_{rot} and $d_{\text{eff}}^{\max}$.

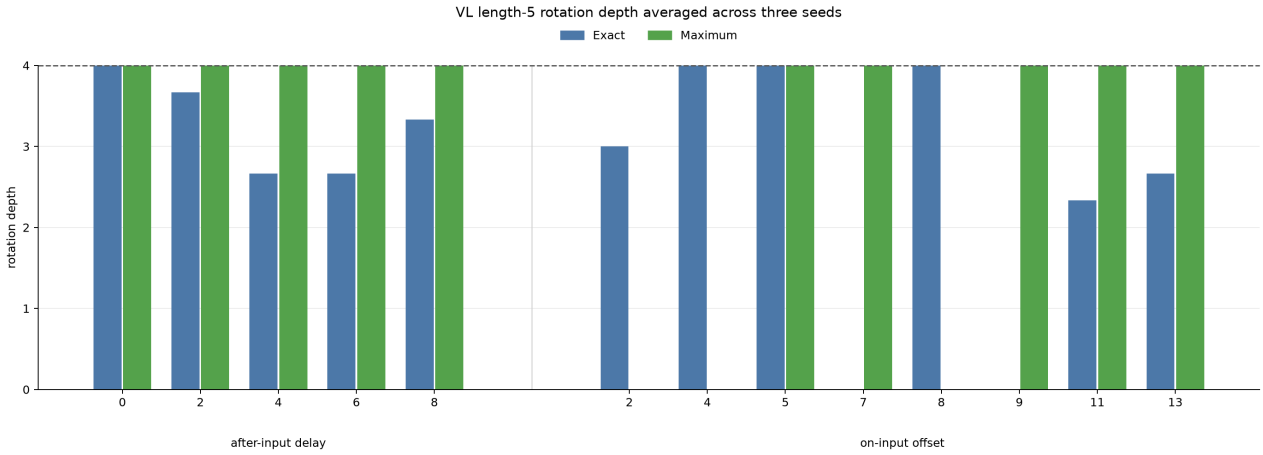


Figure 13: Rotation depths averaged across 3 dataset split and training seeds for runs 53-118 of $d = 512$, $l = 5$ (blue) and $l \leq 5$ (green), and various after and on-input delay values. Missing bars did not have matching runs.

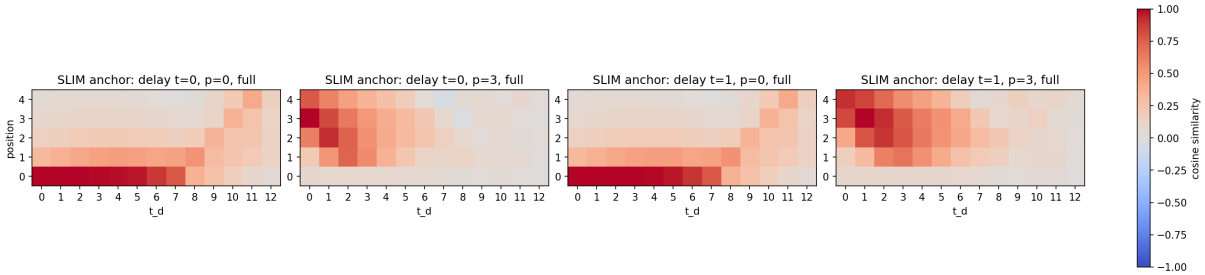


Figure 14: Cosine similarities between phoneme representations for various positions and delay period time steps and anchors for an $l \leq 5$, on-input $m = 13$ run. Notably, rotation does not occur for most delay time steps, thus invalidating our hypothesis that the effective dimensionality must be a function of T.

train exact and maximum length 5 models with various on-input delay values. The on-input offset guarantees a known and constant T for both exact and maximum length trials. However, we observe that maximum length trials generally attained full rotation depths for delay lengths where exact length trials did not.

This is despite $(m+1) \times 41$ exceeding the hidden state dimensionality 512 for $m = 13$, and generally the prediction about full rotations for $\mathbf{h} \in \mathbb{R}^{\geq T \times 41}$ did not hold true. We observed that for delay periods where no input was received, rotations occurred only nearing output time steps with the representations otherwise relatively constant during the delay (Fig. 14).

Ultimately, exact sequence length and thus knowing how long to hold a phoneme for in memory is not the sole pre-requisite for full rotations. Alternatively, it may be that the full rotation depth of on-input delay trials as having benefitted from having, largely, a constant number of phonemes represented in its hidden state which it must persistently cycle through during output. For a delay of m aligned with input start and a sequence length l, there are $\max(l-m, 0)$ out of l time steps during which the number of phonemes represented is constant at L; the greater m, the less time steps in which this occurs. However, this does not account for why maximum length 5 runs consistently attain full rotation depth, despite having samples with a lower l. As such, we fail to reconcile why on-input offsets work consistently for longer sequence lengths (10) but not shorter ones (5). Later, we identify that exact length training samples induce deeper rotation geometries than maximum length samples under some conditions but vice versa for others (Fig. 16).

Hypothesis 3: Perfect rotations occur when the delay period is used to reorganize the phoneme sequence for output readiness, a behavior induced by diverse task samples.

Following the previous experiment, we now attempt to understand why maximum length runs produce consistent rotations for longer delay periods, a trend we will continue to notice (Fig. 16). Furthermore, we identify that the delay period hidden state, which largely presents as fixed slots (Fig. 14) may dictate output behavior. We thus theorize that task diversity forces greater organization in delay period hidden states, with more clearly defined and single-rotation transformable subspaces that yield deeper rotations. In particular, in monkeys, sequence reversal tasks induced the working memory to correctly reorder the sequence for output during the delay period [9]. We hypothesize that longer delay periods favored maximum over exact sequence length trials for developing deeper rotations because they allowed more time for rotation-ready reorganization discussed prior.

We train four variants model of different task diversity levels at different hidden state sizes $256, 320, 384, 416, 448, 480,$ and 512 and delay lengths $0, 2,$ and 4:

1. exact sequence length 7 ($l = 7$) samples, forward recall only
2. $l = 7$, forward and reverse recall
3. maximum sequence length 7 ($l \leq 7$), forward recall only
4. $l \leq 7$, forward and reverse recall

The reverse recall task requires recalling the input sequence in reverse during the go period. Pre-output preparation should order both forward and reverse sequence in output order, which we hypothesize may simultaneously be more rotation-ready. If our hypothesis is true, the decline in rotation depth as delay period length increases should be less for models with diverse training tasks. Furthermore, if longer delay periods were advantageous to output-ready organization, then diverse tasks should have greater rotation depths than their non-diverse counterparts when configurations are otherwise the same.

Ultimately, we find that diverse task settings out-perform their less-diverse counterparts in very limited settings (Fig. 15). We do not observe a greater rotation depth for maximum compared to exact length runs for the same conditions overall ($p = 0.948$, one-sided paired t-test), but do for delay $m = 4$ ($p = 0.011$, one-sided paired t-test). For whether forward and reverse have deeper rotations compared to forward-only runs for the same conditions, statistical significance is observed only for delay $m = 0$, while the longer delay periods show no significance ($p = 0.0040$, one-sided paired t-test).

This violates our hypothesis, which postulates that the difference would be greater for longer delay periods, during which the more diverse model hidden states would attain greater rotation readiness. However, as qualified prior, rotation depth is correlated with rotation-ready pre-output hidden state representations (Fig. 9). Qualitatively, delay

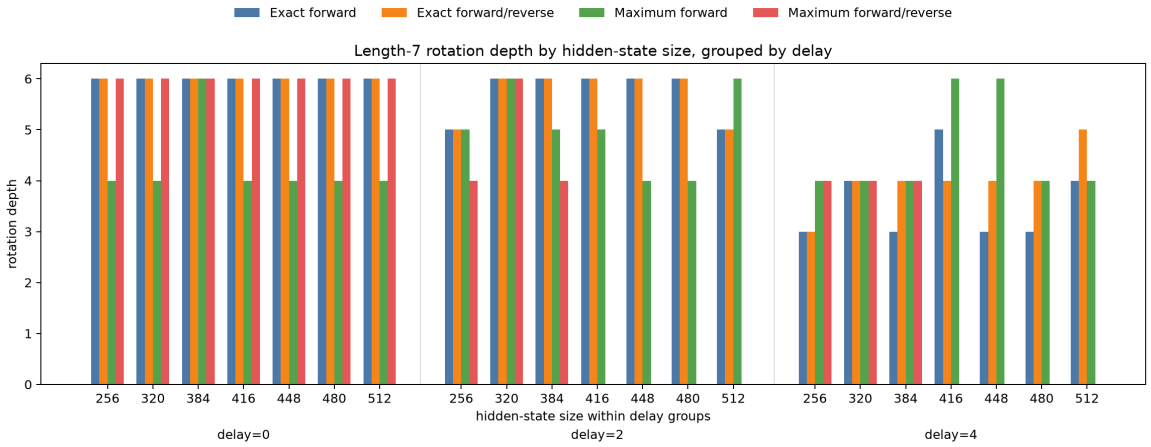


Figure 15: Rotation depths for runs 119-211 with $l = 7$ (exact) and $l \leq 7$ (maximum), forward recall only and forward and reverse recall training tasks, and various hidden state sizes and delay period lengths. Single seed and trial per bar. Missing bars did not have corresponding runs.

$m = 2$ frequently yielded the greatest token representation transformations during the delay period, while that for delay $m = 4$ were near-zero (Fig. 7). In conclusion, these results do not support our hypothesis that task diversity strictly induces greater output-ready reorganization.

Conclusion

The trend observed for $l = 7, l \leq 7$ varieties (Fig. 15) seem inconsistent with those previously observed for $l = 5, l \leq 5$ varieties (Fig. 13), with $l = 7$ attaining greater rotation depths than $l \leq 7$ for shorter delay periods while $l = 5$ consistently attained lesser rotation depths than $l \leq 5$ for all delay period lengths. Thus, we sought to "bridge" these observations by developing a consistent comparison with various l and m values (Fig. 16).

We observe that the hidden state size does not qualitatively affect rotation depth. We observe that exact sequence lengths ($l = N$) outperform maximum sequence lengths ($l \leq N$) for shorter delay periods, but the reverse is true for longer delay periods (Figs. 16,17). We ask whether this is as the longer delay period elicits better rotation-readiness as previously theorized in hypothesis 3. As such, we compute the $\mathbf{T}$ fitting R^2 score (common-map score) on the varying m, l runs (Fig. 17). While a significant correlation is identified between rotation depth and common-map score for exact length runs ($p = 4.4 \times 10^{-14}$ overall, though $p = 0.092$ is not significant when sampling only from $l = 8, l \leq 8$ and $d = 256$), with $\mathbf{T}$ fit declining as the delay period lengthened, no significant relationship over delay period length was observed with maximum length runs. This thus indicates that the delay period doesn't enable maximum length runs to develop better organization; rather, it induces worse organization in exact length runs.

We suspect that this may not hold if we sampled subspaces from earlier delay time steps to fit $\mathbf{T}$. In exact length runs, we observe that strong movement ("rotation") of the first few phonemes occurs in the later part of the delay period (Figs. 7,14). This may emerge particularly in exact length runs as the output start time is consistent across all samples, allowing the model to prepare in advance. Then, this may disrupt whether a common map $\mathbf{T}$ can be fit onto the latest delay hidden state. Thus, we hesitate to conclude that this discrepancy in rotation-readiness between maximum and exact length runs indicates a substantial paradigm difference in delay period organization.

Furthermore, it appears that rotation depth is sensitive to sequence length so much as that rotation depth adjusts semi-proportionally to the overall l, but some truncation affects also appear for lower sequence lengths (Fig. 16). Again, we also observe that exact sequence lengths with no delay period attain perfect rotations, consistent with our previous observations about on-input delays.



Figure 16: Rotation depths for runs 212-499, averaged across 3 different seeds for dataset splits and training for each l, m, d condition. Boxed bars represent runs with PER > 0.01, with maximum PER around 0.04.

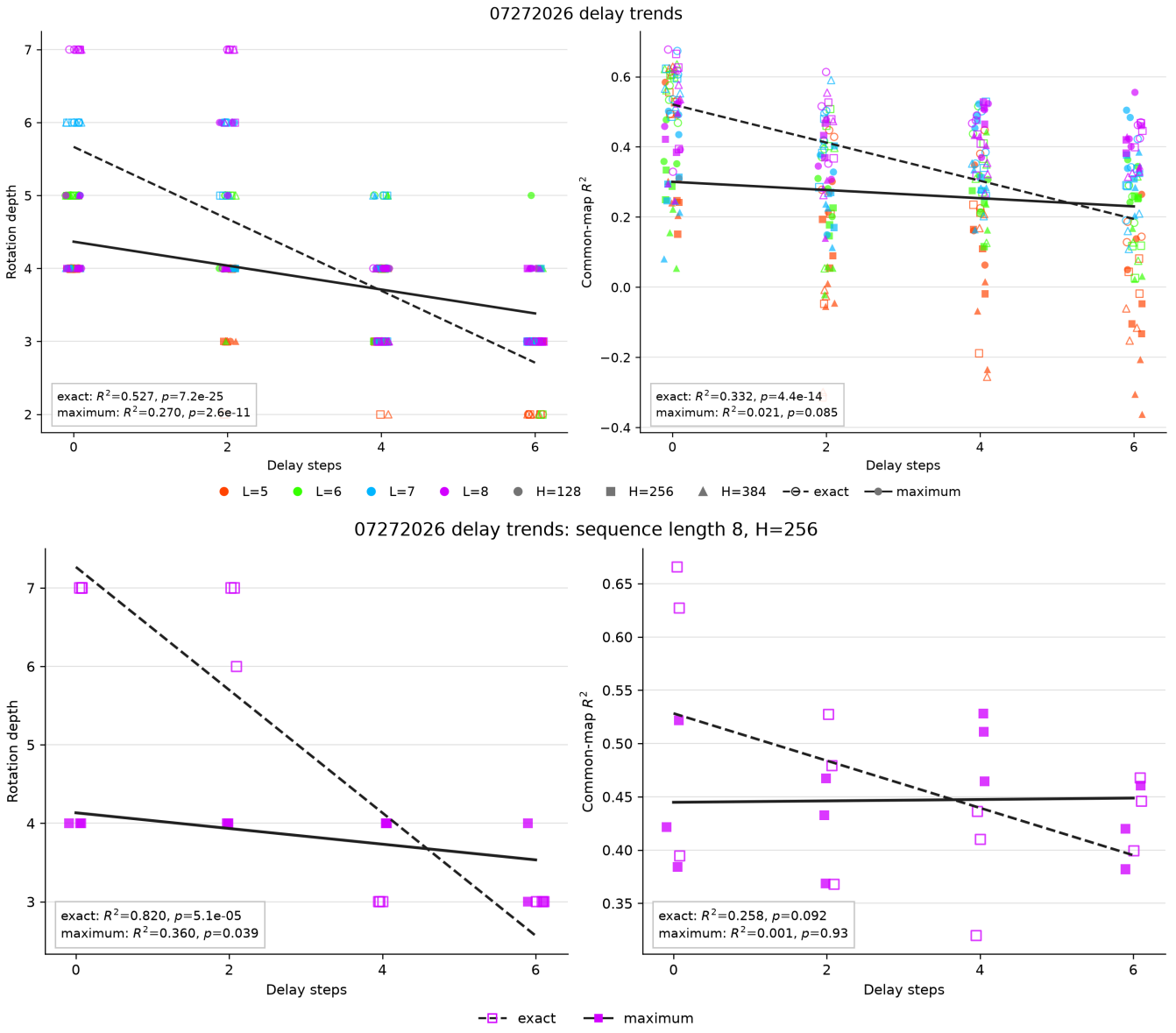


Figure 17: Rotation depth π_{rot} (left) and $\mathbf{T}$ common-map R^2 score (right) versus delay period length m for runs 212-499 of various hidden sequence lengths $l = N, l \leq N$ for $N = 5, 6, 7, 8$ and $d = 128, 256, 384$ (top) and $l = N, l \leq N$ and $d = 256$ only (bottom). Significant correlations were identified between rotation depth and delay length, while only exact length models ($l = N, l = 8$) show significant correlations between common-map score and delay length. Common-map scores for maximum length runs remained relatively static for various delay lengths.

Perfectly rotational representations may just be the exception

From these observations, we conclude that partial and full rotation are not determined by a singular task or training parameter, but on a systematic level by perturbations in delay period and sequence length diversity. Rotation depth is both a function of pre-output hidden state organization, but also constrained by the support for sequence input and hold computations. Furthermore, the perfect subspace rotations observed in previous literature [1], we find, is largely confined to streaming settings. Instead, most of sequence recall only employs rotational subspaces for sequences close to articulation, consistent with previous observations made in the group and in other literature [10]. In longer, naturalistic sequences, tokens are instead stored in slow-moving or fixed slots until close to expression—an observation that is largely universal for nonzero delay periods, memory sizes, regularization parameters, and scross both simple and diverse task settings. Thus, we conclude that speech sequence geometry is, naturally, not perfectly rotating or fully stationary, but an accelerating queue through independently regulated, overlapping subspace trajectories.

References

1. Yang, B. & Yu, S. Orthogonal-Rotational Dynamics Supports Efficient Encoding and Updating for Streaming Information in Working Memory. *Journal of Neuroscience* **46.** ISSN: 0270-6474, 1529-2401. (2026) (Jan. 2026).
2. Gwilliams, L., King, J.-R., Marantz, A. & Poeppel, D. Neural Dynamics of Phoneme Sequences Reveal Position-Invariant Code for Content and Order. *Nature Communications* **13,** 6606. ISSN: 2041-1723. (2026) (Nov. 2022).
3. Penedo, G. *et al. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale* in *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track* (Nov. 2024). (2026).
4. *The CMU Pronouncing Dictionary* http://www.speech.cs.cmu.edu/cgi-bin/cmudict. (2026).
5. Park Kyubyong & Kim, J. *g2pE* 2019.
6. Churchland, M. M. & Shenoy, K. V. Preparatory Activity and the Expansive Null-Space. *Nature reviews. Neuroscience* **25,** 213–236. ISSN: 1471-003X. (2026) (Apr. 2024).
7. Chen, J., Zhang, C., Hu, P., Min, B. & Wang, L. Flexible Control of Sequence Working Memory in the Macaque Frontal Cortex. *Neuron* **112,** 3502–3514.e6. ISSN: 0896-6273. (2026) (Oct. 2024).
8. , W. J. & Fusi, S. Abstract Representations Emerge Naturally in Neural Networks Trained to Perform Multiple Tasks. *Nature Communications* **14,** 1040. ISSN: 2041-1723. (2026) (Feb. 2023).
9. Xie, Y. *et al.* Geometry of Sequence Working Memory in Macaque Prefrontal Cortex. *Science* **375,** 632–639. ISSN: 0036-8075, 1095-9203. (2026) (Feb. 2022).
10. Zhang, M. *et al. From Thought to Action: How a Hierarchy of Neural Dynamics Supports Language Production* Feb. 2025. arXiv: 2502.07429 [q-bio.NC]. (2026).

Table of Runs

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	Common map R^2
1	07/22/26	0/0/0 deterministic	max 10		after input	10		512		1e-4	20	0	0.001	145	14.5	4	partial	0.532	0.772	
2	07/22/26	0/0/0 deterministic	max 10		after input	10		448		1e-4	20	0	0.001	129	12.9	4	partial	0.534	0.778	
3	07/22/26	0/0/0 deterministic	max 10		after input	10		512	384	1e-4	20	0	0.828	6	0.6					
4	07/22/26	0/0/0 deterministic	max 10		after input	10		384		1e-4	20	0	0.001	122	12.2	4	partial	0.549	0.777	
5	07/22/26	0/0/0 deterministic	max 10	within word	after input	10		256		1e-4	20	0	0.002	102	10.2	4	partial	0.588	0.751	
6	07/22/26	0/0/0 deterministic	max 10		after input	10		320		1e-4	20	0	0.001	112	11.2	4	partial	0.534	0.79	
7	07/22/26	0/0/0 deterministic	max 10	full	after input	10		256		1e-4	20	0	0.003	109	10.9	4	partial	0.49	0.717	
8	07/22/26	0/0/0 deterministic	max 10		after input	10		256		1e-4	20	0	0.002	94	9.4	4	partial	0.574	0.794	
9	07/22/26	0/0/0 deterministic	max 10	random	after input	10		256		1e-4	20	0	0.003	116	11.6	4	partial	0.496	0.682	
10	07/22/26	0/0/0 deterministic	max 10		after input	10		128		1e-4	20	0	0.012	63	6.3	4	partial	0.497	0.704	
11	07/22/26	0/0/0 deterministic	max 10	random	after input	10		128		1e-4	20	0	0.018	58	5.8	4	partial	0.227	0.453	
12	07/22/26	0/0/0 deterministic	max 10		after input	10		64		1e-4	20	0	0.176	30	3	8	partial	0.444	0.922	
13	07/22/26	0/0/0 deterministic	max 10	within word	after input	10		64		1e-4	20	0	0.291	28	2.8	4	partial	0.602	0.693	
14	07/22/26	0/0/0 deterministic	max 10	full	after input	10		64		1e-4	20	0	0.408	29	2.9	5	partial	0.639	0.819	
15	07/22/26	0/0/0 deterministic	max 10	random	after input	10		64		1e-4	20	0	0.236	29	2.9	5	partial	0.261	0.523	
16	07/22/26	0/0/0 deterministic	max 10		on input	2		512		1e-4	20	0	0	74	24.667	3	full	-0.034		
17	07/22/26	0/0/0 deterministic	max 10		on input	4		512		1e-4	20	0	0	59	11.8	5	full	0.019		
18	07/22/26	0/0/0 deterministic	max 10		on input	6		512		1e-4	20	0	0	85	12.143	7	full	0.201		
19	07/22/26	0/0/0 deterministic	max 10		on input	8		512		1e-4	20	0	0	113	12.556	9	full	0.403		
20	07/22/26	0/0/0 deterministic	max 20		after input	10		1024		1e-3	20	0	0.011	328	16.4	3	partial	0.377	0.485	
21	07/22/26	0/0/0 deterministic	max 20		after input	10		256		1e-3	20	0	0.037	129	6.45	3	partial	0.412	0.466	
22	07/22/26	0/0/0 deterministic	max 10	within word	after input	10		512		1e-4	20	0	0.001	154	15.4	3	partial	0.477	0.735	
23	07/22/26	0/0/0 deterministic	max 10	full	after input	10		512		1e-4	20	0	0.002	177	17.7	3	partial	0.458	0.719	
24	07/22/26	0/0/0 deterministic	max 10	random	after input	10		512		1e-4	20	0	0.001	181	18.1	3	partial	0.381	0.606	
25	07/22/26	0/0/0 deterministic	max 20		after input	10		512		1e-4	20	0	0.006	211	10.55	2	partial	0.182	0.379	
26	07/22/26	0/0/0 deterministic	max 30		after input	10		512		1e-4	20	0	0.034	257	8.567	2	partial	0.131	0.209	
27	07/22/26	0/0/0 deterministic	max 40		after input	10		512		1e-4	20	0	0.062	269	6.725	2	partial	0.123	0.238	
28	07/22/26	0/0/0 deterministic	max 50		after input	10		512		1e-4	20	0	0.119	275	5.5	2	partial	0.098	0.222	
29	07/22/26	0/0/0 deterministic	max 10		after input	10		512		0e+0	20	0	0.001	145	14.5	4	partial	0.518	0.754	
30	07/22/26	0/0/0 deterministic	max 10		after input	10		512		1e-3	20	0	0.001	138	13.8	4	partial	0.534	0.786	
31	07/22/26	0/0/0 deterministic	max 10		after input	10		512		1e-2	20	0	0	133	13.3	4	partial	0.558	0.779	
32	07/22/26	0/0/0 deterministic	max 10		after input	10		512		1e-1	20	0	0	124	12.4	4	partial	0.537	0.594	
33	07/23/26	0/0/0 deterministic	max 10		after input	3		512		1e-4	20	0	0.001	136	13.6	4	partial	0.505	0.733	
34	07/23/26	0/0/0 deterministic	max 10		after input	5		512		1e-4	20	0	0.001	138	13.8	4	partial	0.509	0.73	
35	07/23/26	0/0/0 deterministic	max 10		after input	7		512		1e-4	20	0	0.001	138	13.8	4	partial	0.531	0.775	
36	07/23/26	0/0/0 deterministic	max 10		after input	9		512		1e-4	20	0	0	147	14.7	3	partial	0.435	0.757	
37	07/23/26	0/0/0 deterministic	max 10		after input	0		512		1e-4	20	0	0.524	271	27.1	1	partial		0.87	
38	07/23/26	0/0/0 deterministic	max 10		on input	10		512		1e-4	20	0	0.001	140	12.727	9	partial	0.492		
39	07/23/26	0/0/0 deterministic	exact 5		on input	2		512		1e-4	20	0	0	85	28.333	3	full	-0.018		
40	07/23/26	0/0/0 deterministic	max 10		on input	20		512		1e-4	20	0	0.001	146	6.952	4	partial	0.528	0.776	
41	07/23/26	0/0/0 deterministic	exact 5		on input	4		512		1e-4	20	0	0	88	17.6	4	full	0.207		
42	07/23/26	0/0/0 deterministic	exact 5		on input	5		512		1e-4	20	0	0	102	17	4	partial	0.315		
43	07/23/26	0/0/0 deterministic	max 10		on input	15		512		1e-4	20	0	0.001	146	9.125	4	partial	0.476	0.754	
44	07/23/26	0/0/0 deterministic	exact 5		on input	8		512		1e-4	20	0	0.001	115	12.778	4	partial	0.363		
45	07/23/26	0/0/0 deterministic	exact 5		on input	11		512		1e-4	20	0	0	124	10.333	3	partial	0.294	0.456	
46	07/23/26	0/0/0 deterministic	exact 5		on input	13		512		1e-4	20	0	0.002	126	9	2	partial	0.22	0.524	
47	07/23/26	0/0/0 deterministic	max 7		after input	10		512		1e-4	20	0	0	120	17.143	4	partial	0.488	0.749	
48	07/23/26	0/0/0 deterministic	exact 5		after input	10		512		1e-4	20	0	0.001	124	24.8	3	partial	0.32	0.584	
49	07/23/26	0/0/0 deterministic	max 5		after input	10		512		1e-4	20	0	0	120	24	3	partial	0.333	0.633	
50	07/23/26	0/0/0 deterministic	max 9		after input	10		512		1e-4	20	0	0	129	14.333	4	partial	0.509	0.758	
51	07/23/26	0/0/0 deterministic	exact 15		after input	10		512		1e-4	20	0	0.007	173	11.533	3	partial	0.468	0.713	
52	07/23/26	0/0/0 deterministic	exact 10		after input	10		512		1e-4	20	0	0.002	143	14.3	3	partial	0.449	0.749	
53	07/24/26	1/1/1 nondeterministic	max 5		on input	4		512		1e-4	20	0	0	83	16.6	4	full	0.153		
54	07/24/26	1/1/1 nondeterministic	exact 5		on input	2		512		1e-4	20	0	0	85	28.333	3	full	-0.027		
55	07/24/26	1/1/1 nondeterministic	max 5		on input	2		512		1e-4	20	0	0	80	26.667	3	full	-0.021		
56	07/24/26	1/1/1 nondeterministic	exact 5		on input	4		512		1e-4	20	0	0	93	18.6	4	full	0.27		
57	07/24/26	1/1/1 nondeterministic	exact 5		on input	5		512		1e-4	20	0	0	105	17.5	4	partial	0.311		
58	07/24/26	2/2/2 nondeterministic	max 5		on input	2		512		1e-4	20	0	0	82	27.333	3	full	-0.023		

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
59	07/24/26	1/1/1 nondeterministic	exact 5		on input	8		512		1e-4	20	0	0	122	13.556	4	partial	0.397		
60	07/24/26	1/1/1 nondeterministic	max 5		on input	8		512		1e-4	20	0	0.001	104	11.556	4	partial	0.34		
61	07/24/26	2/2/2 nondeterministic	exact 5		on input	2		512		1e-4	20	0	0	83	27.667	3	full	-0.012		
62	07/24/26	1/1/1 nondeterministic	exact 5		on input	13		512		1e-4	20	0	0	119	8.5	3	partial	0.301	0.601	
63	07/24/26	2/2/2 nondeterministic	max 5		on input	4		512		1e-4	20	0	0	77	15.4	4	full	0.132		
64	07/24/26	1/1/1 nondeterministic	exact 5		on input	11		512		1e-4	20	0	0	126	10.5	2	partial	0.225	0.538	
65	07/24/26	2/2/2 nondeterministic	exact 5		on input	4		512		1e-4	20	0	0	89	17.8	4	full	0.251		
66	07/24/26	2/2/2 nondeterministic	exact 5		on input	5		512		1e-4	20	0	0	99	16.5	4	partial	0.346		
67	07/24/26	2/2/2 nondeterministic	exact 5		on input	11		512		1e-4	20	0	0	129	10.75	2	partial	0.251	0.537	
68	07/24/26	2/2/2 nondeterministic	exact 5		on input	8		512		1e-4	20	0	0.001	103	11.444	4	partial	0.375		
69	07/24/26	3/3/3 nondeterministic	max 5		on input	2		512		1e-4	20	0	0	78	26	3	full	-0.007		
70	07/24/26	2/2/2 nondeterministic	max 5		on input	8		512		1e-4	20	0	0.001	98	10.889	4	partial	0.342		
71	07/24/26	2/2/2 nondeterministic	exact 5		on input	13		512		1e-4	20	0	0	112	8	3	partial	0.309	0.598	
72	07/24/26	3/3/3 nondeterministic	max 5		on input	4		512		1e-4	20	0	0	77	15.4	4	full	0.144		
73	07/24/26	3/3/3 nondeterministic	max 5		on input	8		512		1e-4	20	0	0.001	107	11.889	4	partial	0.361		
74	07/24/26	1/1/1 nondeterministic	max 5		on input	5		512		1e-4	20	0	0	101	16.833	4	partial	0.281		
75	07/24/26	1/1/1 nondeterministic	max 5		on input	11		512		1e-4	20	0	0	120	10	4	partial	0.319		
76	07/24/26	1/1/1 nondeterministic	max 5		on input	13		512		1e-4	20	0	0.003	103	7.357	4	partial	0.3		
77	07/24/26	2/2/2 nondeterministic	max 5		on input	5		512		1e-4	20	0	0	94	15.667	4	partial	0.284		
78	07/24/26	2/2/2 nondeterministic	max 5		on input	11		512		1e-4	20	0	0.001	109	9.083	4	partial	0.342		
79	07/24/26	2/2/2 nondeterministic	max 5		on input	13		512		1e-4	20	0	0.001	117	8.357	4	partial	0.358		
80	07/24/26	3/3/3 nondeterministic	max 5		on input	5		512		1e-4	20	0	0	92	15.333	4	partial	0.281		
81	07/24/26	3/3/3 nondeterministic	max 5		on input	11		512		1e-4	20	0	0.001	109	9.083	4	partial	0.346		
82	07/24/26	3/3/3 nondeterministic	max 5		on input	13		512		1e-4	20	0	0	101	7.214	4	partial	0.317		
83	07/24/26	1/1/1 nondeterministic	exact 5		after input	0		512		1e-4	20	0	0	105	21	4	full	0.311		
84	07/24/26	2/2/2 nondeterministic	exact 5		after input	0		512		1e-4	20	0	0	99	19.8	4	full	0.346		
85	07/24/26	3/3/3 nondeterministic	exact 5		after input	0		512		1e-4	20	0	0	96	19.2	4	full	0.297		
86	07/24/26	3/3/3 nondeterministic	exact 5		after input	2		512		1e-4	20	0	0.001	113	22.6	3	partial	0.326	0.588	
87	07/24/26	2/2/2 nondeterministic	exact 5		after input	2		512		1e-4	20	0	0	111	22.2	4	full	0.38		
88	07/24/26	1/1/1 nondeterministic	exact 5		after input	2		512		1e-4	20	0	0	122	24.4	4	full	0.36		
89	07/24/26	1/1/1 nondeterministic	exact 5		after input	4		512		1e-4	20	0	0	131	26.2	2	partial	0.231	0.542	
90	07/24/26	2/2/2 nondeterministic	exact 5		after input	4		512		1e-4	20	0	0	124	24.8	3	partial	0.295	0.53	
91	07/24/26	1/1/1 nondeterministic	exact 5		after input	6		512		1e-4	20	0	0	126	25.2	2	partial	0.225	0.538	
92	07/24/26	3/3/3 nondeterministic	exact 5		after input	4		512		1e-4	20	0	0	122	24.4	3	partial	0.304	0.563	
93	07/24/26	2/2/2 nondeterministic	exact 5		after input	6		512		1e-4	20	0	0	123	24.6	3	partial	0.307	0.578	
94	07/24/26	3/3/3 nondeterministic	exact 5		after input	6		512		1e-4	20	0	0	122	24.4	3	partial	0.283	0.476	
95	07/24/26	3/3/3 nondeterministic	exact 5		after input	8		512		1e-4	20	0	0.001	116	23.2	4	full	0.387		
96	07/24/26	2/2/2 nondeterministic	exact 5		after input	8		512		1e-4	20	0	0	112	22.4	3	partial	0.309	0.598	
97	07/24/26	1/1/1 nondeterministic	exact 5		after input	8		512		1e-4	20	0	0.001	120	24	3	partial	0.307	0.552	
98	07/24/26	1/1/1 nondeterministic	max 5		after input	0		512		1e-4	20	0	0.001	112	22.4	4	full	0.175		
99	07/24/26	2/2/2 nondeterministic	max 5		after input	0		512		1e-4	20	0	0	110	22	4	full	0.2		
100	07/24/26	3/3/3 nondeterministic	max 5		after input	0		512		1e-4	20	0	0	116	23.2	4	full	0.303		
101	07/24/26	1/1/1 nondeterministic	max 5		after input	2		512		1e-4	20	0	0.002	125	25	4	full	0.355		
102	07/24/26	2/2/2 nondeterministic	max 5		after input	2		512		1e-4	20	0	0.001	116	23.2	4	full	0.349		
103	07/24/26	1/1/1 nondeterministic	max 5		after input	4		512		1e-4	20	0	0.003	118	23.6	4	full	0.376		
104	07/24/26	3/3/3 nondeterministic	max 5		after input	2		512		1e-4	20	0	0	113	22.6	4	full	0.333		
105	07/24/26	1/1/1 nondeterministic	max 5		after input	6		512		1e-4	20	0	0.008	129	25.8	4	full	0.382		
106	07/24/26	2/2/2 nondeterministic	max 5		after input	4		512		1e-4	20	0	0.001	115	23	4	full	0.36		
107	07/24/26	3/3/3 nondeterministic	max 5		after input	4		512		1e-4	20	0	0	102	20.4	4	full	0.332		
108	07/24/26	2/2/2 nondeterministic	max 5		after input	6		512		1e-4	20	0	0.001	120	24	4	full	0.375		
109	07/24/26	3/3/3 nondeterministic	max 5		after input	6		512		1e-4	20	0	0	118	23.6	4	full	0.342		
110	07/24/26	1/1/1 nondeterministic	max 5		on input	7		512		1e-4	20	0	0.002	118	14.75	4	partial	0.364		
111	07/24/26	3/3/3 nondeterministic	max 5		after input	8		512		1e-4	20	0	0.001	106	21.2	4	full	0.374		
112	07/24/26	2/2/2 nondeterministic	max 5		on input	7		512		1e-4	20	0	0	100	12.5	4	partial	0.349		
113	07/24/26	1/1/1 nondeterministic	max 5		after input	8		512		1e-4	20	0	0.001	119	23.8	4	full	0.364		
114	07/24/26	3/3/3 nondeterministic	max 5		on input	7		512		1e-4	20	0	0	101	12.625	4	partial	0.323		
115	07/24/26	2/2/2 nondeterministic	max 5		after input	8		512		1e-4	20	0	0.001	115	23	4	full	0.389		
116	07/24/26	2/2/2 nondeterministic	max 5		on input	9		512		1e-4	20	0	0	112	11.2	4	partial	0.359		
117	07/24/26	1/1/1 nondeterministic	max 5		on input	9		512		1e-4	20	0	0	112	11.2	4	partial	0.346		
118	07/24/26	3/3/3 nondeterministic	max 5		on input	9		512		1e-4	20	0	0	106	10.6	4	partial	0.312		
119	07/25/26	0/0/0 deterministic	exact 7		after input	0		480		1e-4	20	0	0							0.517

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	τ_{rot}	Ext.	s_{rot}	s_{fixed}	**T** R^2
120	07/25/26	0/0/0 deterministic	exact 7		after input	0		448		1e-4	20	0	0							0.39
121	07/25/26	0/0/0 deterministic	exact 7		after input	0		512	480	1e-4	20	0	0.001							
122	07/25/26	0/0/0 deterministic	exact 7		after input	0		384		1e-4	20	0	0							0.444
123	07/25/26	0/0/0 deterministic	exact 7		after input	0		416		1e-4	20	0	0							0.302
124	07/25/26	0/0/0 deterministic	exact 7		after input	0		512		1e-4	20	0	0							0.476
125	07/25/26	0/0/0 deterministic	exact 7		after input	0		512	448	1e-4	20	0	0.007							
126	07/25/26	0/0/0 deterministic	exact 7		after input	0		512	416	1e-4	20	0	0.005							
127	07/25/26	0/0/0 deterministic	exact 7		after input	0		512	384	1e-4	20	0	0.005							
128	07/25/26	0/0/0 deterministic	exact 7		after input	2		448		1e-4	20	0	0							
129	07/25/26	0/0/0 deterministic	exact 7		after input	2		512		1e-4	20	0	0.001							
130	07/25/26	0/0/0 deterministic	exact 7		after input	2		480		1e-4	20	0	0							0.384
131	07/25/26	0/0/0 deterministic	exact 7		after input	2		384		1e-4	20	0	0							0.456
132	07/25/26	0/0/0 deterministic	exact 7		after input	2		416		1e-4	20	0	0							0.391
133	07/25/26	0/0/0 deterministic	exact 7		after input	2		512	416	1e-4	20	0	0.026							
134	07/25/26	0/0/0 deterministic	exact 7		after input	2		512	384	1e-4	20	0	0.031							
135	07/25/26	0/0/0 deterministic	exact 7		after input	2		512	448	1e-4	20	0	0.05							
136	07/25/26	0/0/0 deterministic	exact 7		after input	4		512		1e-4	20	0	0							0.408
137	07/25/26	0/0/0 deterministic	exact 7		after input	4		448		1e-4	20	0	0							0.16
138	07/25/26	0/0/0 deterministic	exact 7		after input	4		416		1e-4	20	0	0							0.308
139	07/25/26	0/0/0 deterministic	exact 7		after input	4		512	384	1e-4	20	0	0.821							
140	07/25/26	0/0/0 deterministic	exact 7		after input	4		480		1e-4	20	0	0							0.209
141	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	512		1e-4	20	0	0							0.517
142	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	448		1e-4	20	0	0							0.443
143	07/25/26	0/0/0 deterministic	exact 7		after input	4		384		1e-4	20	0	0.001							0.306
144	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	480		1e-4	20	0	0							0.332
145	07/25/26	0/0/0 deterministic	exact 7		after input	4		512	416	1e-4	20	0	0.092							
146	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	416		1e-4	20	0	0							0.48
147	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	512	480	1e-4	20	0	0.019							
148	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	384		1e-4	20	0	0							0.415
149	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	512	384	1e-4	20	0	0.013							
150	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	512	416	1e-4	20	0	0.011							
151	07/25/26	0/0/0 deterministic	exact 7		after input	0	true	512	448	1e-4	20	0	0.01							
152	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	512		1e-4	20	0	0							0.498
153	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	480		1e-4	20	0	0							0.508
154	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	416		1e-4	20	0	0							0.484
155	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	384		1e-4	20	0	0							0.547
156	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	448		1e-4	20	0	0							0.525
157	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	512	480	1e-4	20	0	0.159							
158	07/25/26	0/0/0 deterministic	exact 7		after input	2	true	512	416	1e-4	20	0	0.059							
159	07/25/26	0/0/0 deterministic	exact 7		after input	4	true	512		1e-4	20	0	0.001							0.4
160	07/25/26	0/0/0 deterministic	exact 7		after input	4	true	480		1e-4	20	0	0							0.338
161	07/25/26	0/0/0 deterministic	exact 7		after input	4	true	448		1e-4	20	0	0							0.29
162	07/25/26	0/0/0 deterministic	exact 7		after input	4	true	416		1e-4	20	0	0.001							0.062
163	07/25/26	0/0/0 deterministic	exact 7		after input	4	true	384		1e-4	20	0	0.001							0.277
164	07/25/26	0/0/0 deterministic	max 7		after input	0		512		1e-4	20	0	0							0.22
165	07/25/26	0/0/0 deterministic	max 7		after input	0		448		1e-4	20	0	0							0.08
166	07/25/26	0/0/0 deterministic	max 7		after input	0		416		1e-4	20	0	0							0.286
167	07/25/26	0/0/0 deterministic	max 7		after input	0		384		1e-4	20	0	0							0.139
168	07/25/26	0/0/0 deterministic	max 7		after input	0		480		1e-4	20	0	0							0.371
169	07/25/26	0/0/0 deterministic	max 7		after input	2		480		1e-4	20	0	0							0.168
170	07/25/26	0/0/0 deterministic	max 7		after input	2		512		1e-4	20	0	0							0.22
171	07/25/26	0/0/0 deterministic	max 7		after input	0		512	384	1e-4	20	0	0.097							
172	07/25/26	0/0/0 deterministic	max 7		after input	2		416		1e-4	20	0	0							0.234
173	07/25/26	0/0/0 deterministic	max 7		after input	2		384		1e-4	20	0	0							0.08
174	07/25/26	0/0/0 deterministic	max 7		after input	2		448		1e-4	20	0	0							0.208
175	07/25/26	0/0/0 deterministic	max 7		after input	4		448		1e-4	20	0	0							0.299
176	07/25/26	0/0/0 deterministic	max 7		after input	4		480		1e-4	20	0	0							0.168
177	07/25/26	0/0/0 deterministic	max 7		after input	4		512		1e-4	20	0	0							0.232
178	07/25/26	0/0/0 deterministic	max 7		after input	4		416		1e-4	20	0	0							-0.025
179	07/25/26	0/0/0 deterministic	max 7		after input	4		384		1e-4	20	0	0.001							0.293
180	07/25/26	0/0/0 deterministic	max 7		after input	0	true	512		1e-4	20	0	0.001							-0.072

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
181	07/25/26	0/0/0 deterministic	max 7		after input	0	true	480		1e-4	20	0	0							0.086
182	07/25/26	0/0/0 deterministic	max 7		after input	0	true	448		1e-4	20	0	0							0.245
183	07/25/26	0/0/0 deterministic	max 7		after input	0	true	416		1e-4	20	0	0							0.261
184	07/25/26	0/0/0 deterministic	max 7		after input	2	true	512		1e-4	20	0	0							
185	07/26/26	0/0/0 deterministic	exact 7		after input	2		320		1e-4	20	0	0.001							0.609
186	07/26/26	0/0/0 deterministic	exact 7		after input	2		256		1e-4	20	0	0.001							
187	07/26/26	0/0/0 deterministic	exact 7		after input	0		256		1e-4	20	0	0							0.377
188	07/26/26	0/0/0 deterministic	exact 7		after input	4		320		1e-4	20	0	0.001							0.258
189	07/26/26	0/0/0 deterministic	exact 7		after input	0		320		1e-4	20	0	0							0.293
190	07/26/26	0/0/0 deterministic	exact 7		after input	2	true	320		1e-4	20	0	0							0.396
191	07/26/26	0/0/0 deterministic	exact 7		after input	0	true	320		1e-4	20	0	0							0.27
192	07/26/26	0/0/0 deterministic	exact 7		after input	0	true	256		1e-4	20	0	0.001							0.419
193	07/26/26	0/0/0 deterministic	exact 7		after input	4	true	320		1e-4	20	0	0.001							0.348
194	07/26/26	0/0/0 deterministic	max 7		after input	0		320		1e-4	20	0	0							0.343
195	07/26/26	0/0/0 deterministic	exact 7		after input	4	true	256		1e-4	20	0	0							-0.044
196	07/26/26	0/0/0 deterministic	exact 7		after input	2	true	256		1e-4	20	0	0.001							0.478
197	07/26/26	0/0/0 deterministic	max 7		after input	0		256		1e-4	20	0	0							0.063
198	07/26/26	0/0/0 deterministic	exact 7		after input	4		256		1e-4	20	0	0							0.171
199	07/26/26	0/0/0 deterministic	max 7		after input	2		320		1e-4	20	0	0.001							0.242
200	07/26/26	0/0/0 deterministic	max 7		after input	2		256		1e-4	20	0	0.001							0.253
201	07/26/26	0/0/0 deterministic	max 7		after input	4		256		1e-4	20	0	0.001							0.504
202	07/26/26	0/0/0 deterministic	max 7		after input	4		320		1e-4	20	0	0							0.421
203	07/26/26	0/0/0 deterministic	max 7		after input	0	true	384		1e-4	20	0	0							0.305
204	07/26/26	0/0/0 deterministic	max 7		after input	2	true	384		1e-4	20	0	0							
205	07/26/26	0/0/0 deterministic	max 7		after input	0	true	256		1e-4	20	0	0							0.343
206	07/26/26	0/0/0 deterministic	max 7		after input	2	true	320		1e-4	20	0	0.001							
207	07/26/26	0/0/0 deterministic	max 7		after input	0	true	320		1e-4	20	0	0.001							-0.012
208	07/26/26	0/0/0 deterministic	max 7		after input	4	true	384		1e-4	20	0	0							
209	07/26/26	0/0/0 deterministic	max 7		after input	4	true	320		1e-4	20	0	0.001							
210	07/26/26	0/0/0 deterministic	max 7		after input	2	true	256		1e-4	20	0	0							
211	07/26/26	0/0/0 deterministic	max 7		after input	4	true	256		1e-4	20	0	0.001							
212	07/27/26	1/1/1 deterministic	exact 5		after input	0		128		1e-4	10	0.001	0.002	70	14	4	full	0.384		0.613
213	07/27/26	2/2/2 deterministic	exact 5		after input	0		128		1e-4	10	0.001	0.002	66	13.2	4	full	0.327		0.494
214	07/27/26	3/3/3 deterministic	exact 5		after input	0		256		1e-4	10	0.001	0.001	95	19	4	full	0.36		0.528
215	07/27/26	2/2/2 deterministic	exact 5		after input	0		256		1e-4	10	0.001	0	94	18.8	4	full	0.371		0.494
216	07/27/26	1/1/1 deterministic	exact 5		after input	0		256		1e-4	10	0.001	0.001	93	18.6	4	full	0.378		0.556
217	07/27/26	2/2/2 deterministic	exact 5		after input	0		384		1e-4	10	0.001	0	106	21.2	4	full	0.367		0.606
218	07/27/26	1/1/1 deterministic	exact 5		after input	0		384		1e-4	10	0.001	0.001	114	22.8	4	full	0.366		0.629
219	07/27/26	3/3/3 deterministic	exact 5		after input	0		384		1e-4	10	0.001	0	114	22.8	4	full	0.352		0.624
220	07/27/26	3/3/3 deterministic	exact 5		after input	0		128		1e-4	10	0.001	0.003	67	13.4	4	full	0.342		0.533
221	07/27/26	2/2/2 deterministic	exact 5		after input	2		256		1e-4	10	0.001	0	95	19	2	partial	0.328	0.572	-0.314
222	07/27/26	3/3/3 deterministic	exact 5		after input	2		256		1e-4	10	0.001	0.004	100	20	4	full	0.425		0.21
223	07/27/26	1/1/1 deterministic	exact 5		after input	2		384		1e-4	10	0.001	0.002	112	22.4	2	partial	0.312	0.452	-0.299
224	07/27/26	1/1/1 deterministic	exact 5		after input	2		256		1e-4	10	0.001	0.002	99	19.8	2	partial	0.341	0.528	-0.047
225	07/27/26	3/3/3 deterministic	exact 5		after input	2		384		1e-4	10	0.001	0.002	115	23	3	partial	0.347	0.531	-0.008
226	07/27/26	2/2/2 deterministic	exact 5		after input	2		384		1e-4	10	0.001	0	114	22.8	3	partial	0.345	0.51	-0.026
227	07/27/26	1/1/1 deterministic	exact 5		after input	2		128		1e-4	10	0.001	0.016	66	13.2	4	full	0.375		0.428
228	07/27/26	2/2/2 deterministic	exact 5		after input	2		128		1e-4	10	0.001	0.011	67	13.4	4	full	0.417		0.278
229	07/27/26	1/1/1 deterministic	exact 5		after input	4		384		1e-4	10	0.001	0.002	109	21.8	2	partial	0.302	0.484	-0.256
230	07/27/26	1/1/1 deterministic	exact 5		after input	4		256		1e-4	10	0.001	0.004	95	19	4	full	0.415		0.213
231	07/27/26	2/2/2 deterministic	exact 5		after input	4		128		1e-4	10	0.001	0.009	64	12.8	4	full	0.41		0.448
232	07/27/26	2/2/2 deterministic	exact 5		after input	4		256		1e-4	10	0.001	0.001	92	18.4	2	partial	0.325	0.598	-0.188
233	07/27/26	3/3/3 deterministic	exact 5		after input	4		384		1e-4	10	0.001	0.002	115	23	4	full	0.423		0.168
234	07/27/26	3/3/3 deterministic	exact 5		after input	4		128		1e-4	10	0.001	0.027	65	13	4	full	0.37		0.38
235	07/27/26	2/2/2 deterministic	exact 5		after input	4		384		1e-4	10	0.001	0.001	109	21.8	3	partial	0.357	0.601	0.208
236	07/27/26	1/1/1 deterministic	exact 5		after input	4		128		1e-4	10	0.001	0.016	67	13.4	3	partial	0.36	0.495	0.225
237	07/27/26	3/3/3 deterministic	exact 5		after input	4		256		1e-4	10	0.001	0.013	97	19.4	3	partial	0.423	0.636	0.235
238	07/27/26	3/3/3 deterministic	exact 5		after input	2		128		1e-4	10	0.001	0.015	66	13.2	4	full	0.443		0.447
239	07/27/26	3/3/3 deterministic	exact 6		after input	0		256		1e-4	10	0.001	0	86	14.333	5	full	0.358		0.598
240	07/27/26	1/1/1 deterministic	exact 6		after input	0		256		1e-4	10	0.001	0.001	86	14.333	5	full	0.385		0.604
241	07/27/26	2/2/2 deterministic	exact 6		after input	2		256		1e-4	10	0.001	0.002	99	16.5	5	full	0.43		0.46

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
242	07/27/26	1/1/1 deterministic	exact 6		after input	0		128		1e-4	10	0.001	0.002	65	10.833	5	full	0.383		0.575
243	07/27/26	1/1/1 deterministic	exact 6		after input	4		256		1e-4	10	0.001	0.002	97	16.167	4	partial	0.467	0.683	0.366
244	07/27/26	1/1/1 deterministic	exact 6		after input	2		256		1e-4	10	0.001	0.001	93	15.5	5	full	0.418		0.508
245	07/27/26	3/3/3 deterministic	exact 6		after input	2		256		1e-4	10	0.001	0.001	97	16.167	5	full	0.423		0.403
246	07/27/26	2/2/2 deterministic	exact 6		after input	0		256		1e-4	10	0.001	0	89	14.833	5	full	0.403		0.608
247	07/27/26	3/3/3 deterministic	exact 6		after input	6		256		1e-4	10	0.001	0.003	97	16.167	2	partial	0.367	0.633	0.119
248	07/27/26	1/1/1 deterministic	exact 7		after input	0		256		1e-4	10	0.001	0	83	11.857	6	full	0.365		0.624
249	07/27/26	2/2/2 deterministic	exact 7		after input	0		256		1e-4	10	0.001	0.001	88	12.571	6	full	0.378		0.595
250	07/27/26	2/2/2 deterministic	exact 7		after input	2		256		1e-4	10	0.001	0.002	104	14.857	5	partial	0.442	0.635	0.285
251	07/27/26	3/3/3 deterministic	exact 6		after input	4		256		1e-4	10	0.001	0.002	99	16.5	3	partial	0.468	0.664	0.271
252	07/27/26	2/2/2 deterministic	exact 6		after input	4		256		1e-4	10	0.001	0.001	96	16	5	full	0.415		0.41
253	07/27/26	2/2/2 deterministic	exact 6		after input	6		256		1e-4	10	0.001	0.003	99	16.5	3	partial	0.458	0.638	0.027
254	07/27/26	1/1/1 deterministic	exact 6		after input	6		256		1e-4	10	0.001	0.002	98	16.333	3	partial	0.448	0.681	0.117
255	07/27/26	1/1/1 deterministic	exact 5		after input	6		128		1e-4	10	0.001	0.026	66	13.2	2	partial	0.286	0.42	0.186
256	07/27/26	1/1/1 deterministic	exact 6		after input	0		384		1e-4	10	0.001	0	106	17.667	5	full	0.39		0.635
257	07/27/26	1/1/1 deterministic	exact 7		after input	4		256		1e-4	10	0.001	0.002	98	14	5	partial	0.498	0.723	0.529
258	07/27/26	1/1/1 deterministic	exact 7		after input	2		256		1e-4	10	0.001	0.001	95	13.571	6	full	0.499		0.481
259	07/27/26	2/2/2 deterministic	exact 5		after input	6		128		1e-4	10	0.001	0.012	63	12.6	2	partial	0.289	0.544	0.144
260	07/27/26	2/2/2 deterministic	exact 7		after input	4		256		1e-4	10	0.001	0.001	99	14.143	3	partial	0.47	0.657	0.341
261	07/27/26	1/1/1 deterministic	exact 7		after input	6		256		1e-4	10	0.001	0.002	97	13.857	3	partial	0.474	0.676	0.288
262	07/27/26	3/3/3 deterministic	exact 7		after input	6		256		1e-4	10	0.001	0.003	101	14.429	3	partial	0.475	0.678	0.291
263	07/27/26	1/1/1 deterministic	exact 8		after input	0		256		1e-4	10	0.001	0.001	94	11.75	7	full	0.453		0.627
264	07/27/26	3/3/3 deterministic	exact 8		after input	0		256		1e-4	10	0.001	0.001	80	10	7	full	0.347		0.395
265	07/27/26	1/1/1 deterministic	exact 6		after input	2		128		1e-4	10	0.001	0.01	65	10.833	5	full	0.413		0.394
266	07/27/26	2/2/2 deterministic	exact 8		after input	0		256		1e-4	10	0.001	0	90	11.25	7	full	0.445		0.666
267	07/27/26	3/3/3 deterministic	exact 6		after input	0		128		1e-4	10	0.001	0.002	61	10.167	5	full	0.371		0.534
268	07/27/26	2/2/2 deterministic	exact 8		after input	2		256		1e-4	10	0.001	0.001	97	12.125	7	full	0.518		0.527
269	07/27/26	2/2/2 deterministic	exact 6		after input	2		128		1e-4	10	0.001	0.014	66	11	4	partial	0.446	0.64	0.371
270	07/27/26	2/2/2 deterministic	exact 7		after input	6		256		1e-4	10	0.001	0.002	99	14.143	3	partial	0.477	0.645	0.338
271	07/27/26	1/1/1 deterministic	exact 6		after input	4		128		1e-4	10	0.001	0.014	64	10.667	3	partial	0.416	0.611	0.211
272	07/27/26	1/1/1 deterministic	exact 8		after input	2		256		1e-4	10	0.001	0.002	100	12.5	6	partial	0.512	0.597	0.368
273	07/27/26	3/3/3 deterministic	exact 6		after input	2		128		1e-4	10	0.001	0.012	68	11.333	5	full	0.412		0.416
274	07/27/26	1/1/1 deterministic	exact 8		after input	4		256		1e-4	10	0.001	0.005	105	13.125	3	partial	0.515	0.738	0.436
275	07/27/26	3/3/3 deterministic	exact 8		after input	2		256		1e-4	10	0.001	0.003	95	11.875	7	full	0.425		0.479
276	07/27/26	2/2/2 deterministic	exact 6		after input	4		128		1e-4	10	0.001	0.013	63	10.5	4	partial	0.481	0.684	0.516
277	07/27/26	2/2/2 deterministic	exact 8		after input	4		256		1e-4	10	0.001	0.002	100	12.5	3	partial	0.483	0.665	0.41
278	07/27/26	2/2/2 deterministic	exact 6		after input	0		128		1e-4	10	0.001	0.003	62	10.333	5	full	0.308		0.468
279	07/27/26	2/2/2 deterministic	exact 6		after input	0		384		1e-4	10	0.001	0.001	101	16.833	5	full	0.376		0.561
280	07/27/26	3/3/3 deterministic	exact 6		after input	4		128		1e-4	10	0.001	0.012	64	10.667	5	full	0.47		0.437
281	07/27/26	1/1/1 deterministic	max 6		after input	0		256		1e-4	10	0.001	0.001	104	17.333	5	full	0.518		0.249
282	07/27/26	3/3/3 deterministic	exact 8		after input	6		256		1e-4	10	0.001	0.006	104	13	3	partial	0.51	0.755	0.446
283	07/27/26	1/1/1 deterministic	max 6		after input	2		256		1e-4	10	0.001	0.001	100	16.667	4	partial	0.451	0.743	0.178
284	07/27/26	3/3/3 deterministic	max 6		after input	0		256		1e-4	10	0.001	0.001	103	17.167	5	full	0.498		0.334
285	07/27/26	3/3/3 deterministic	exact 8		after input	4		256		1e-4	10	0.001	0.003	104	13	3	partial	0.488	0.712	0.32
286	07/27/26	2/2/2 deterministic	exact 6		after input	6		128		1e-4	10	0.001	0.024	65	10.833	3	partial	0.444	0.637	0.29
287	07/27/26	1/1/1 deterministic	exact 6		after input	2		384		1e-4	10	0.001	0.001	125	20.833	4	partial	0.428	0.568	0.054
288	07/27/26	1/1/1 deterministic	exact 8		after input	6		256		1e-4	10	0.001	0.003	104	13	3	partial	0.511	0.725	0.399
289	07/27/26	1/1/1 deterministic	exact 6		after input	6		128		1e-4	10	0.001	0.02	63	10.5	3	partial	0.403	0.57	0.258
290	07/27/26	2/2/2 deterministic	exact 6		after input	2		384		1e-4	10	0.001	0	120	20	5	full	0.427		0.397
291	07/27/26	3/3/3 deterministic	exact 6		after input	0		384		1e-4	10	0.001	0.001	106	17.667	5	full	0.354		0.622
292	07/27/26	2/2/2 deterministic	max 6		after input	2		256		1e-4	10	0.001	0	103	17.167	4	partial	0.455	0.748	0.226
293	07/27/26	2/2/2 deterministic	exact 8		after input	6		256		1e-4	10	0.001	0.003	103	12.875	3	partial	0.533	0.75	0.468
294	07/27/26	2/2/2 deterministic	max 6		after input	0		256		1e-4	10	0.001	0.001	103	17.167	4	partial	0.517	0.702	0.287
295	07/27/26	3/3/3 deterministic	max 6		after input	2		256		1e-4	10	0.001	0.002	102	17	4	partial	0.472	0.706	0.146
296	07/27/26	2/2/2 deterministic	max 6		after input	4		256		1e-4	10	0.001	0.001	99	16.5	3	partial	0.426	0.704	0.115
297	07/27/26	3/3/3 deterministic	max 6		after input	4		256		1e-4	10	0.001	0.002	101	16.833	3	partial	0.443	0.69	0.275
298	07/27/26	3/3/3 deterministic	exact 6		after input	2		384		1e-4	10	0.001	0.001	113	18.833	5	full	0.374		0.48
299	07/27/26	1/1/1 deterministic	exact 6		after input	4		384		1e-4	10	0.001	0.002	122	20.333	3	partial	0.435	0.589	0.126
300	07/27/26	2/2/2 deterministic	exact 6		after input	4		384		1e-4	10	0.001	0.002	121	20.167	4	partial	0.459	0.69	0.309
301	07/27/26	3/3/3 deterministic	exact 6		after input	4		384		1e-4	10	0.001	0.002	113	18.833	5	full	0.415		0.35
302	07/27/26	1/1/1 deterministic	max 6		after input	4		256		1e-4	10	0.001	0.002	96	16	4	partial	0.514	0.733	0.241

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
303	07/27/26	2/2/2 deterministic	max 6		after input	6		256		1e-4	10	0.001	0.002	100	16.667	3	partial	0.471	0.722	0.256
304	07/27/26	1/1/1 deterministic	exact 6		after input	6		384		1e-4	10	0.001	0.002	121	20.167	2	partial	0.37	0.601	0.075
305	07/27/26	3/3/3 deterministic	max 6		after input	6		256		1e-4	10	0.001	0.002	94	15.667	4	partial	0.519	0.792	0.251
306	07/27/26	1/1/1 deterministic	max 6		after input	6		256		1e-4	10	0.001	0.002	96	16	3	partial	0.482	0.713	0.26
307	07/27/26	1/1/1 deterministic	max 7		after input	0		256		1e-4	10	0.001	0	101	14.429	4	partial	0.527	0.712	0.294
308	07/27/26	2/2/2 deterministic	exact 6		after input	6		384		1e-4	10	0.001	0.003	121	20.167	3	partial	0.452	0.645	0.127
309	07/27/26	2/2/2 deterministic	max 7		after input	2		256		1e-4	10	0.001	0	97	13.857	4	partial	0.541	0.768	0.403
310	07/27/26	2/2/2 deterministic	max 7		after input	0		256		1e-4	10	0.001	0.001	108	15.429	4	partial	0.513	0.701	0.314
311	07/27/26	2/2/2 deterministic	max 7		after input	4		256		1e-4	10	0.001	0.001	96	13.714	4	partial	0.499	0.688	0.276
312	07/27/26	1/1/1 deterministic	max 7		after input	4		256		1e-4	10	0.001	0.001	96	13.714	4	partial	0.518	0.747	0.313
313	07/27/26	1/1/1 deterministic	max 8		after input	2		256		1e-4	10	0.001	0.001	98	12.25	4	partial	0.53	0.686	0.369
314	07/27/26	1/1/1 deterministic	max 7		after input	6		256		1e-4	10	0.001	0.001	97	13.857	3	partial	0.495	0.772	0.377
315	07/27/26	2/2/2 deterministic	max 8		after input	0		256		1e-4	10	0.001	0.001	105	13.125	4	partial	0.553	0.753	0.421
316	07/27/26	1/1/1 deterministic	max 7		after input	2		256		1e-4	10	0.001	0.001	104	14.857	4	partial	0.492	0.652	0.17
317	07/27/26	3/3/3 deterministic	max 8		after input	2		256		1e-4	10	0.001	0.001	101	12.625	4	partial	0.518	0.688	0.467
318	07/27/26	3/3/3 deterministic	max 7		after input	6		256		1e-4	10	0.001	0.002	99	14.143	3	partial	0.482	0.752	0.309
319	07/27/26	2/2/2 deterministic	max 7		after input	6		256		1e-4	10	0.001	0.001	94	13.429	4	partial	0.562	0.754	0.324
320	07/27/26	3/3/3 deterministic	max 8		after input	0		256		1e-4	10	0.001	0	100	12.5	4	partial	0.525	0.738	0.384
321	07/27/26	1/1/1 deterministic	max 8		after input	0		256		1e-4	10	0.001	0.001	106	13.25	4	partial	0.58	0.779	0.522
322	07/27/26	2/2/2 deterministic	max 8		after input	2		256		1e-4	10	0.001	0.001	101	12.625	4	partial	0.546	0.78	0.433
323	07/27/26	3/3/3 deterministic	exact 5		after input	6		128		1e-4	10	0.001	0.026	63	12.6	2	partial	0.312	0.491	0.128
324	07/27/26	3/3/3 deterministic	exact 6		after input	6		384		1e-4	10	0.001	0.002	122	20.333	3	partial	0.414	0.612	0.316
325	07/27/26	1/1/1 deterministic	exact 7		after input	0		384		1e-4	10	0.001	0	106	15.143	6	full	0.387		0.553
326	07/27/26	2/2/2 deterministic	exact 7		after input	0		384		1e-4	10	0.001	0	98	14	6	full	0.357		0.566
327	07/27/26	3/3/3 deterministic	exact 6		after input	6		128		1e-4	10	0.001	0.022	64	10.667	3	partial	0.353	0.512	0.183
328	07/27/26	1/1/1 deterministic	max 8		after input	6		256		1e-4	10	0.001	0.001	102	12.75	3	partial	0.545	0.775	0.46
329	07/27/26	1/1/1 deterministic	max 8		after input	4		256		1e-4	10	0.001	0.001	102	12.75	4	partial	0.576	0.792	0.528
330	07/27/26	2/2/2 deterministic	max 8		after input	6		256		1e-4	10	0.001	0.002	100	12.5	3	partial	0.518	0.779	0.42
331	07/27/26	2/2/2 deterministic	max 8		after input	4		256		1e-4	10	0.001	0.001	97	12.125	4	partial	0.559	0.771	0.464
332	07/27/26	3/3/3 deterministic	max 8		after input	6		256		1e-4	10	0.001	0.001	96	12	4	partial	0.554	0.782	0.382
333	07/27/26	1/1/1 deterministic	exact 5		after input	6		256		1e-4	10	0.001	0.003	92	18.4	2	partial	0.344	0.55	0.045
334	07/27/26	2/2/2 deterministic	exact 5		after input	6		256		1e-4	10	0.001	0.002	90	18	2	partial	0.313	0.573	-0.018
335	07/27/26	1/1/1 deterministic	exact 7		after input	0		128		1e-4	10	0.001	0.002	66	9.429	6	full	0.462		0.673
336	07/27/26	2/2/2 deterministic	exact 7		after input	0		128		1e-4	10	0.001	0.004	62	8.857	6	full	0.344		0.535
337	07/27/26	1/1/1 deterministic	exact 7		after input	2		128		1e-4	10	0.001	0.008	65	9.286	6	full	0.473		0.503
338	07/27/26	2/2/2 deterministic	exact 7		after input	2		128		1e-4	10	0.001	0.01	65	9.286	6	full	0.424		0.495
339	07/27/26	1/1/1 deterministic	exact 7		after input	4		128		1e-4	10	0.001	0.014	64	9.143	4	partial	0.415	0.617	0.385
340	07/27/26	3/3/3 deterministic	max 8		after input	4		256		1e-4	10	0.001	0.001	100	12.5	4	partial	0.558	0.805	0.511
341	07/27/26	1/1/1 deterministic	exact 7		after input	6		128		1e-4	10	0.001	0.012	63	9	3	partial	0.376	0.621	0.342
342	07/27/26	2/2/2 deterministic	exact 7		after input	4		128		1e-4	10	0.001	0.014	63	9	5	partial	0.494	0.7	0.501
343	07/27/26	3/3/3 deterministic	exact 7		after input	6		128		1e-4	10	0.001	0.014	65	9.286	3	partial	0.434	0.643	0.422
344	07/27/26	2/2/2 deterministic	exact 8		after input	2		128		1e-4	10	0.001	0.011	57	7.125	7	full	0.328		0.613
345	07/27/26	2/2/2 deterministic	exact 7		after input	6		128		1e-4	10	0.001	0.008	63	9	3	partial	0.381	0.563	0.19
346	07/27/26	1/1/1 deterministic	exact 7		after input	2		384		1e-4	10	0.001	0.001	123	17.571	5	partial	0.48	0.696	0.269
347	07/27/26	1/1/1 deterministic	exact 8		after input	0		128		1e-4	10	0.001	0.003	62	7.75	7	full	0.434		0.677
348	07/27/26	3/3/3 deterministic	exact 5		after input	6		256		1e-4	10	0.001	0.005	89	17.8	2	partial	0.337	0.55	0.082
349	07/27/26	3/3/3 deterministic	exact 8		after input	0		128		1e-4	10	0.001	0.003	62	7.75	7	full	0.403		0.618
350	07/27/26	2/2/2 deterministic	exact 8		after input	0		128		1e-4	10	0.001	0.002	57	7.125	7	full	0.367		0.329
351	07/27/26	1/1/1 deterministic	exact 8		after input	2		128		1e-4	10	0.001	0.012	65	8.125	4	partial	0.57	0.699	0.516
352	07/27/26	3/3/3 deterministic	exact 8		after input	2		128		1e-4	10	0.001	0.01	59	7.375	7	full	0.405		0.352
353	07/27/26	1/1/1 deterministic	exact 8		after input	6		128		1e-4	10	0.001	0.044	62	7.75	3	partial	0.409	0.663	0.47
354	07/27/26	3/3/3 deterministic	max 6		after input	0		128		1e-4	10	0.001	0.002	65	10.833	5	full	0.522		0.477
355	07/27/26	2/2/2 deterministic	exact 7		after input	2		384		1e-4	10	0.001	0	119	17	6	full	0.367		0.59
356	07/27/26	2/2/2 deterministic	max 6		after input	0		128		1e-4	10	0.001	0.002	67	11.167	5	full	0.521		0.352
357	07/27/26	2/2/2 deterministic	exact 8		after input	4		128		1e-4	10	0.001	0.023	65	8.125	4	partial	0.509	0.722	0.475
358	07/27/26	3/3/3 deterministic	exact 8		after input	4		128		1e-4	10	0.001	0.018	66	8.25	3	partial	0.454	0.696	0.489
359	07/27/26	1/1/1 deterministic	max 6		after input	0		128		1e-4	10	0.001	0.002	66	11	5	full	0.51		0.358
360	07/27/26	2/2/2 deterministic	exact 8		after input	6		128		1e-4	10	0.001	0.028	64	8	3	partial	0.47	0.716	0.424
361	07/27/26	1/1/1 deterministic	exact 8		after input	4		128		1e-4	10	0.001	0.024	66	8.25	4	partial	0.52	0.757	0.467
362	07/27/26	1/1/1 deterministic	exact 5		after input	6		384		1e-4	10	0.001	0.002	110	22	2	partial	0.301	0.484	-0.152
363	07/27/26	2/2/2 deterministic	max 6		after input	2		128		1e-4	10	0.001	0.003	65	10.833	4	partial	0.476	0.71	0.31

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
364	07/27/26	1/1/1 deterministic	max 6		after input	2		128		1e-4	10	0.001	0.005	66	11	4	partial	0.474	0.75	0.281
365	07/27/26	3/3/3 deterministic	max 6		after input	2		128		1e-4	10	0.001	0.003	64	10.667	4	partial	0.488	0.752	0.201
366	07/27/26	3/3/3 deterministic	max 6		after input	4		128		1e-4	10	0.001	0.006	64	10.667	4	partial	0.456	0.733	0.214
367	07/27/26	3/3/3 deterministic	exact 8		after input	6		128		1e-4	10	0.001	0.021	65	8.125	3	partial	0.412	0.694	0.462
368	07/27/26	2/2/2 deterministic	max 6		after input	4		128		1e-4	10	0.001	0.006	66	11	3	partial	0.474	0.713	0.319
369	07/27/26	2/2/2 deterministic	max 6		after input	6		128		1e-4	10	0.001	0.008	63	10.5	3	partial	0.47	0.758	0.363
370	07/27/26	1/1/1 deterministic	max 6		after input	6		128		1e-4	10	0.001	0.007	65	10.833	5	full	0.527		0.242
371	07/27/26	1/1/1 deterministic	exact 7		after input	4		384		1e-4	10	0.001	0.001	129	18.429	3	partial	0.479	0.666	0.28
372	07/27/26	1/1/1 deterministic	max 7		after input	2		128		1e-4	10	0.001	0.005	64	9.143	5	partial	0.513	0.693	0.149
373	07/27/26	1/1/1 deterministic	max 6		after input	4		128		1e-4	10	0.001	0.005	61	10.167	4	partial	0.497	0.76	0.307
374	07/27/26	2/2/2 deterministic	max 7		after input	2		128		1e-4	10	0.001	0.004	64	9.143	6	full	0.48		0.328
375	07/27/26	3/3/3 deterministic	max 6		after input	6		128		1e-4	10	0.001	0.007	62	10.333	4	partial	0.486	0.754	0.344
376	07/27/26	1/1/1 deterministic	max 7		after input	0		128		1e-4	10	0.001	0.005	67	9.571	5	partial	0.516	0.662	0.435
377	07/27/26	2/2/2 deterministic	exact 5		after input	6		384		1e-4	10	0.001	0.001	109	21.8	2	partial	0.331	0.528	-0.116
378	07/27/26	1/1/1 deterministic	max 7		after input	4		128		1e-4	10	0.001	0.01	65	9.286	4	partial	0.542	0.776	0.49
379	07/27/26	3/3/3 deterministic	max 7		after input	6		128		1e-4	10	0.001	0.004	64	9.143	3	partial	0.5	0.708	0.505
380	07/27/26	2/2/2 deterministic	max 7		after input	0		128		1e-4	10	0.001	0.003	69	9.857	4	partial	0.556	0.718	0.502
381	07/27/26	2/2/2 deterministic	max 7		after input	4		128		1e-4	10	0.001	0.003	62	8.857	4	partial	0.571	0.754	0.522
382	07/27/26	2/2/2 deterministic	max 7		after input	6		128		1e-4	10	0.001	0.007	63	9	3	partial	0.476	0.749	0.339
383	07/27/26	1/1/1 deterministic	max 8		after input	0		128		1e-4	10	0.001	0.003	66	8.25	5	partial	0.551	0.752	0.527
384	07/27/26	2/2/2 deterministic	max 8		after input	0		128		1e-4	10	0.001	0.005	67	8.375	4	partial	0.502	0.704	0.392
385	07/27/26	2/2/2 deterministic	exact 7		after input	4		384		1e-4	10	0.001	0.001	130	18.571	5	partial	0.498	0.628	0.352
386	07/27/26	2/2/2 deterministic	exact 7		after input	6		384		1e-4	10	0.001	0.001	126	18	3	partial	0.473	0.69	0.109
387	07/27/26	2/2/2 deterministic	max 8		after input	2		128		1e-4	10	0.001	0.006	66	8.25	6	partial	0.459	0.68	0.345
388	07/27/26	1/1/1 deterministic	exact 7		after input	6		384		1e-4	10	0.001	0.001	123	17.571	3	partial	0.473	0.673	0.16
389	07/27/26	3/3/3 deterministic	exact 5		after input	6		384		1e-4	10	0.001	0.004	111	22.2	2	partial	0.302	0.557	-0.061
390	07/27/26	1/1/1 deterministic	max 7		after input	6		128		1e-4	10	0.001	0.015	62	8.857	4	partial	0.585	0.791	0.484
391	07/27/26	1/1/1 deterministic	exact 8		after input	0		384		1e-4	10	0.001	0	100	12.5	7	full	0.4		0.536
392	07/27/26	3/3/3 deterministic	exact 8		after input	0		384		1e-4	10	0.001	0.001	106	13.25	7	full	0.394		0.577
393	07/27/26	2/2/2 deterministic	exact 8		after input	0		384		1e-4	10	0.001	0	109	13.625	7	full	0.431		0.623
394	07/27/26	3/3/3 deterministic	exact 7		after input	6		384		1e-4	10	0.001	0.002	128	18.286	3	partial	0.461	0.617	0.21
395	07/27/26	3/3/3 deterministic	max 8		after input	2		128		1e-4	10	0.001	0.003	63	7.875	6	partial	0.47	0.654	0.307
396	07/27/26	3/3/3 deterministic	max 8		after input	0		128		1e-4	10	0.001	0.002	68	8.5	5	partial	0.532	0.719	0.458
397	07/27/26	2/2/2 deterministic	exact 8		after input	2		384		1e-4	10	0.001	0	120	15	6	partial	0.478	0.677	0.262
398	07/27/26	1/1/1 deterministic	exact 8		after input	2		384		1e-4	10	0.001	0	126	15.75	7	full	0.489		0.473
399	07/27/26	3/3/3 deterministic	max 8		after input	6		128		1e-4	10	0.001	0.009	65	8.125	4	partial	0.515	0.74	0.401
400	07/27/26	1/1/1 deterministic	max 8		after input	2		128		1e-4	10	0.001	0.004	67	8.375	5	partial	0.561	0.711	0.478
401	07/27/26	1/1/1 deterministic	exact 8		after input	4		384		1e-4	10	0.001	0.001	129	16.125	3	partial	0.506	0.709	0.322
402	07/27/26	2/2/2 deterministic	exact 8		after input	4		384		1e-4	10	0.001	0.001	125	15.625	3	partial	0.522	0.719	0.354
403	07/27/26	2/2/2 deterministic	max 8		after input	4		128		1e-4	10	0.001	0.009	66	8.25	4	partial	0.545	0.755	0.523
404	07/27/26	1/1/1 deterministic	max 6		after input	0		384		1e-4	10	0.001	0	121	20.167	5	full	0.479		0.155
405	07/27/26	3/3/3 deterministic	exact 8		after input	2		384		1e-4	10	0.001	0.001	116	14.5	7	full	0.445		0.555
406	07/27/26	1/1/1 deterministic	exact 8		after input	6		384		1e-4	10	0.001	0.001	128	16	3	partial	0.524	0.716	0.314
407	07/27/26	3/3/3 deterministic	exact 8		after input	6		384		1e-4	10	0.001	0.005	130	16.25	3	partial	0.525	0.687	0.329
408	07/27/26	1/1/1 deterministic	max 8		after input	4		128		1e-4	10	0.001	0.005	65	8.125	4	partial	0.55	0.746	0.506
409	07/27/26	3/3/3 deterministic	exact 8		after input	4		384		1e-4	10	0.001	0.002	123	15.375	3	partial	0.48	0.671	0.267
410	07/27/26	2/2/2 deterministic	exact 8		after input	6		384		1e-4	10	0.001	0.001	129	16.125	3	partial	0.494	0.71	0.325
411	07/27/26	0/0/0 deterministic	exact 7		after input	0		128		1e-4	10	0.001	0.002	64	9.143	6	full	0.418		0.606
412	07/27/26	2/2/2 deterministic	max 6		after input	2		384		1e-4	10	0.001	0	124	20.667	3	partial	0.433	0.693	-0.021
413	07/27/26	1/1/1 deterministic	max 8		after input	6		128		1e-4	10	0.001	0.006	64	8	4	partial	0.577	0.793	0.423
414	07/27/26	3/3/3 deterministic	max 6		after input	0		384		1e-4	10	0.001	0	126	21	4	partial	0.508	0.704	0.222
415	07/27/26	3/3/3 deterministic	max 6		after input	2		384		1e-4	10	0.001	0.001	120	20	4	partial	0.46	0.687	0.058
416	07/27/26	1/1/1 deterministic	max 6		after input	4		384		1e-4	10	0.001	0	115	19.167	3	partial	0.41	0.681	0.163
417	07/27/26	2/2/2 deterministic	max 6		after input	0		384		1e-4	10	0.001	0	128	21.333	4	partial	0.489	0.704	0.054
418	07/27/26	1/1/1 deterministic	max 6		after input	2		384		1e-4	10	0.001	0	119	19.833	4	partial	0.449	0.698	0.054
419	07/27/26	1/1/1 deterministic	max 6		after input	6		384		1e-4	10	0.001	0.001	116	19.333	3	partial	0.436	0.693	0.023
420	07/27/26	2/2/2 deterministic	max 6		after input	6		384		1e-4	10	0.001	0	122	20.333	3	partial	0.446	0.707	0.168
421	07/27/26	2/2/2 deterministic	max 6		after input	4		384		1e-4	10	0.001	0	116	19.333	5	full	0.497		0.237
422	07/27/26	3/3/3 deterministic	max 6		after input	6		384		1e-4	10	0.001	0.001	120	20	4	partial	0.487	0.736	0.031
423	07/27/26	2/2/2 deterministic	max 7		after input	0		384		1e-4	10	0.001	0	127	18.143	4	partial	0.512	0.673	0.081
424	07/27/26	3/3/3 deterministic	max 6		after input	4		384		1e-4	10	0.001	0	113	18.833	4	partial	0.445	0.631	0.443

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
425	07/27/26	0/0/0 deterministic	exact 7		after input	0		256		1e-4	10	0.001	0	84	12	6	full	0.367		0.486
426	07/27/26	1/1/1 deterministic	max 7		after input	0		384		1e-4	10	0.001	0	126	18	4	partial	0.503	0.7	0.244
427	07/27/26	0/0/0 deterministic	exact 7		after input	0		384		1e-4	10	0.001	0	102	14.571	6	full	0.368		0.513
428	07/27/26	1/1/1 deterministic	max 7		after input	4		384		1e-4	10	0.001	0.001	115	16.429	4	partial	0.498	0.744	0.283
429	07/27/26	2/2/2 deterministic	max 8		after input	6		128		1e-4	10	0.001	0.006	65	8.125	4	partial	0.569	0.785	0.555
430	07/27/26	2/2/2 deterministic	max 7		after input	4		384		1e-4	10	0.001	0	123	17.571	5	partial	0.506	0.609	0.335
431	07/27/26	1/1/1 deterministic	max 7		after input	6		384		1e-4	10	0.001	0.001	119	17	3	partial	0.471	0.756	0.284
432	07/27/26	0/0/0 deterministic	exact 7		after input	2		384		1e-4	10	0.001	0.001	117	16.714	6	full	0.42		0.408
433	07/27/26	1/1/1 deterministic	max 7		after input	2		384		1e-4	10	0.001	0	127	18.143	4	partial	0.432	0.553	0.226
434	07/27/26	3/3/3 deterministic	max 8		after input	4		128		1e-4	10	0.001	0.006	64	8	4	partial	0.481	0.778	0.37
435	07/27/26	0/0/0 deterministic	exact 7		after input	2		128		1e-4	10	0.001	0.009	61	8.714	6	full	0.385		0.419
436	07/27/26	0/0/0 deterministic	exact 7		after input	2		256		1e-4	10	0.001	0.002	104	14.857	5	partial	0.456	0.598	0.384
437	07/27/26	2/2/2 deterministic	max 7		after input	2		384		1e-4	10	0.001	0	117	16.714	4	partial	0.494	0.656	0.237
438	07/27/26	2/2/2 deterministic	max 7		after input	6		384		1e-4	10	0.001	0.001	124	17.714	3	partial	0.483	0.721	0.204
439	07/27/26	1/1/1 deterministic	max 5		after input	0		128		1e-4	10	0.001	0.004	69	13.8	4	full	0.512		0.584
440	07/27/26	0/0/0 deterministic	exact 7		after input	4		256		1e-4	10	0.001	0.001	103	14.714	3	partial	0.469	0.626	0.26
441	07/27/26	2/2/2 deterministic	max 5		after input	0		128		1e-4	10	0.001	0.003	70	14	4	full	0.456		0.49
442	07/27/26	0/0/0 deterministic	exact 7		after input	4		384		1e-4	10	0.001	0.002	129	18.429	3	partial	0.468	0.672	0.202
443	07/27/26	3/3/3 deterministic	max 5		after input	0		128		1e-4	10	0.001	0.003	71	14.2	4	full	0.489		0.31
444	07/27/26	0/0/0 deterministic	exact 7		after input	4		128		1e-4	10	0.001	0.018	67	9.571	3	partial	0.484	0.681	0.353
445	07/27/26	1/1/1 deterministic	max 5		after input	0		256		1e-4	10	0.001	0.002	107	21.4	4	full	0.484		0.241
446	07/27/26	2/2/2 deterministic	max 5		after input	0		256		1e-4	10	0.001	0	103	20.6	4	full	0.479		0.247
447	07/27/26	1/1/1 deterministic	max 8		after input	0		384		1e-4	10	0.001	0	129	16.125	4	partial	0.526	0.665	0.294
448	07/27/26	2/2/2 deterministic	max 8		after input	2		384		1e-4	10	0.001	0	122	15.25	4	partial	0.495	0.664	0.139
449	07/27/26	3/3/3 deterministic	max 8		after input	2		384		1e-4	10	0.001	0	122	15.25	4	partial	0.462	0.598	0.282
450	07/27/26	3/3/3 deterministic	max 5		after input	0		256		1e-4	10	0.001	0.001	107	21.4	4	full	0.5		0.151
451	07/27/26	1/1/1 deterministic	max 8		after input	4		384		1e-4	10	0.001	0	119	14.875	3	partial	0.471	0.732	0.404
452	07/27/26	1/1/1 deterministic	max 8		after input	2		384		1e-4	10	0.001	0	122	15.25	4	partial	0.478	0.629	0.305
453	07/27/26	1/1/1 deterministic	max 5		after input	0		384		1e-4	10	0.001	0	129	25.8	4	full	0.466		0.302
454	07/27/26	2/2/2 deterministic	max 5		after input	0		384		1e-4	10	0.001	0	123	24.6	4	full	0.453		0.24
455	07/27/26	3/3/3 deterministic	max 5		after input	0		384		1e-4	10	0.001	0	123	24.6	4	full	0.462		0.204
456	07/27/26	3/3/3 deterministic	max 7		after input	6		384		1e-4	10	0.001	0.001	113	16.143	4	partial	0.547	0.753	0.385
457	07/27/26	2/2/2 deterministic	max 8		after input	0		384		1e-4	10	0.001	0	136	17	5	partial	0.534	0.531	0.497
458	07/27/26	3/3/3 deterministic	max 8		after input	0		384		1e-4	10	0.001	0	128	16	4	partial	0.53	0.709	0.246
459	07/27/26	2/2/2 deterministic	max 5		after input	2		256		1e-4	10	0.001	0.002	101	20.2	4	full	0.455		0.089
460	07/27/26	1/1/1 deterministic	max 5		after input	2		256		1e-4	10	0.001	0.003	100	20	3	partial	0.449	0.709	0.193
461	07/27/26	3/3/3 deterministic	max 8		after input	4		384		1e-4	10	0.001	0	122	15.25	4	partial	0.546	0.746	0.432
462	07/27/26	3/3/3 deterministic	max 5		after input	2		256		1e-4	10	0.001	0.002	102	20.4	4	full	0.435		0.054
463	07/27/26	2/2/2 deterministic	max 5		after input	2		128		1e-4	10	0.001	0.005	66	13.2	3	partial	0.449	0.694	0.27
464	07/27/26	1/1/1 deterministic	max 5		after input	2		384		1e-4	10	0.001	0.002	121	24.2	3	partial	0.391	0.739	-0.045
465	07/27/26	2/2/2 deterministic	max 8		after input	4		384		1e-4	10	0.001	0	120	15	3	partial	0.491	0.751	0.431
466	07/27/26	1/1/1 deterministic	max 8		after input	6		384		1e-4	10	0.001	0	125	15.625	4	partial	0.56	0.766	0.34
467	07/27/26	2/2/2 deterministic	max 5		after input	2		384		1e-4	10	0.001	0.001	120	24	4	full	0.406		0.01
468	07/27/26	3/3/3 deterministic	max 5		after input	2		128		1e-4	10	0.001	0.008	63	12.6	4	full	0.458		0.213
469	07/27/26	1/1/1 deterministic	max 5		after input	2		128		1e-4	10	0.001	0.007	67	13.4	4	full	0.434		0.301
470	07/27/26	3/3/3 deterministic	max 5		after input	2		384		1e-4	10	0.001	0	121	24.2	4	full	0.425		-0.055
471	07/27/26	1/1/1 deterministic	max 5		after input	4		256		1e-4	10	0.001	0.008	96	19.2	3	partial	0.398	0.693	0.165
472	07/27/26	2/2/2 deterministic	max 8		after input	6		384		1e-4	10	0.001	0.001	123	15.375	3	partial	0.469	0.775	0.345
473	07/27/26	1/1/1 deterministic	max 5		after input	4		384		1e-4	10	0.001	0.004	120	24	4	full	0.385		-0.235
474	07/27/26	3/3/3 deterministic	max 5		after input	4		384		1e-4	10	0.001	0.001	115	23	3	partial	0.398	0.686	0.015
475	07/27/26	3/3/3 deterministic	max 8		after input	6		384		1e-4	10	0.001	0.001	120	15	4	partial	0.527	0.748	0.429
476	07/27/26	3/3/3 deterministic	max 5		after input	4		256		1e-4	10	0.001	0.002	94	18.8	3	partial	0.385	0.699	0.109
477	07/27/26	1/1/1 deterministic	max 5		after input	4		128		1e-4	10	0.001	0.009	64	12.8	3	partial	0.384	0.738	0.348
478	07/27/26	1/1/1 deterministic	max 5		after input	6		384		1e-4	10	0.001	0.005	117	23.4	3	partial	0.38	0.674	-0.305
479	07/27/26	2/2/2 deterministic	max 5		after input	4		256		1e-4	10	0.001	0.002	98	19.6	3	partial	0.385	0.719	-0.019
480	07/27/26	2/2/2 deterministic	max 5		after input	4		384		1e-4	10	0.001	0.001	115	23	3	partial	0.411	0.708	-0.068
481	07/27/26	2/2/2 deterministic	max 5		after input	6		256		1e-4	10	0.001	0.002	93	18.6	3	partial	0.434	0.713	-0.133
482	07/27/26	2/2/2 deterministic	max 5		after input	6		384		1e-4	10	0.001	0.001	113	22.6	3	partial	0.395	0.748	-0.206
483	07/27/26	3/3/3 deterministic	max 5		after input	6		384		1e-4	10	0.001	0.002	117	23.4	3	partial	0.382	0.649	-0.362
484	07/27/26	1/1/1 deterministic	max 5		after input	6		256		1e-4	10	0.001	0.005	92	18.4	3	partial	0.378	0.665	-0.105
485	07/27/26	3/3/3 deterministic	max 5		after input	6		256		1e-4	10	0.001	0.004	94	18.8	3	partial	0.413	0.756	-0.047

Continued on next page

No.	Date	Seeds	Seq.	Scram.	Delay	Steps	Rev.	H	Rank	WD	Pat.	Δ	PER	$d_{\text{eff}}^{\max}$	$d_{\text{eff}}^{\max}/L$	π_{rot}	Ext.	s_{rot}	s_{fixed}	$\mathbf{T}$ R^2
486	07/27/26	1/1/1 deterministic	max 5		after input	6		128		1e-4	10	0.001	0.014	65	13	3	partial	0.394	0.705	0.05
487	07/27/26	2/2/2 deterministic	max 5		after input	4		128		1e-4	10	0.001	0.005	64	12.8	4	full	0.505		0.063
488	07/27/26	2/2/2 deterministic	max 5		after input	6		128		1e-4	10	0.001	0.01	61	12.2	3	partial	0.415	0.685	0.265
489	07/27/26	3/3/3 deterministic	max 5		after input	4		128		1e-4	10	0.001	0.012	63	12.6	3	partial	0.392	0.65	0.161
490	07/27/26	3/3/3 deterministic	max 5		after input	6		128		1e-4	10	0.001	0.011	63	12.6	3	partial	0.418	0.705	0.138
491	07/27/26	0/0/0 deterministic	max 7		after input	0		384		1e-4	10	0.001	0	130	18.571	4	partial	0.5	0.658	0.213
492	07/27/26	0/0/0 deterministic	max 7		after input	4		384		1e-4	10	0.001	0	123	17.571	3	partial	0.444	0.66	0.164
493	07/27/26	0/0/0 deterministic	max 7		after input	2		256		1e-4	10	0.001	0.001	98	14	4	partial	0.473	0.677	0.27
494	07/27/26	0/0/0 deterministic	max 7		after input	2		384		1e-4	10	0.001	0.001	125	17.857	4	partial	0.468	0.598	0.113
495	07/27/26	0/0/0 deterministic	max 7		after input	4		256		1e-4	10	0.001	0.001	98	14	3	partial	0.508	0.747	0.475
496	07/27/26	0/0/0 deterministic	max 7		after input	0		256		1e-4	10	0.001	0.001	103	14.714	4	partial	0.523	0.715	0.246
497	07/27/26	0/0/0 deterministic	max 7		after input	0		128		1e-4	10	0.001	0.002	67	9.571	5	partial	0.568	0.781	0.491
498	07/27/26	0/0/0 deterministic	max 7		after input	4		128		1e-4	10	0.001	0.006	65	9.286	4	partial	0.578	0.766	0.454
499	07/27/26	0/0/0 deterministic	max 7		after input	2		128		1e-4	10	0.001	0.005	65	9.286	6	full	0.516		0.377